MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

**Project** **Mansourah Massarah Gold Project**
**Customer:** **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

**Title:** **CALCULATION**
**3650 POX ACID WASH,ELTN. & CARBON REGN.**
**DESIGN OF TANK FOUNDATION**

**Document Number:** **MD-216-3650-EG-CV-CAL-1341**
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## DEFINITIONS AND ABBREVIATIONS

| | | |
|---|---|---|
| Document | : | Information and the medium on which it is contained (ISO 9000:2015) |
| EMPLOYER | : | 'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives |
| Contractor | : | Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract). |
| 'PMT' | : | Ma'aden Project Management Team |
| 'MMGP Project' | : | Mansourah Massarah Gold Processing Project |
| Attribute | : | Information element that describes a document or tag. |
| Supplier | : | An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor |
| Sub-Supplier | : | A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor. |
| QA | : | Quality Assurance |
| QC | : | Quality Control |
| Works | : | Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities |
| ELTN. | : | Elution |
| REGN. | : | Regeneration |


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## 1. INTENT

This document is to prepare design of foundation for **3610-TNK-001,3650-COL-001 & 002,3650-TNK-001,003 & 006 and 3820-TNK-001**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

## 2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

## 3. SCOPE

The analysis has been carried out in Excel and Staad. For member strength calculation force values are directly taken from Load Data Provided & Output of Staad analysis as applicable.


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# 4. DESIGN REQUIREMENTS

## 4.1 Design Load

All Design Loads are indicated in Input drawings attached in Attachment - 1. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

# 5. CONCRETE FOUNDATIONS

## 5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

## 5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)


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## 5.3 Minimum Concrete Cover

| Concrete Exposure | Member | Minimum Cover(mm) |
|---|---|---|
| Exposed to weather or contact with ground | Foundation | 50 |
| Exposed to weather or contact with ground | Column | 50 |

## 5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 375 Kn/$m^2$ at 1.2 m below FGL. ( Refer Table 5 - BH-11 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).


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## 6. DESIGN CALCULATION FOR FOUNDATION


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**DESIGN OF FOUNDATION 3610-TNK-001**

| | Along-X | Along-Z |
|---|---|---|
| Size of column : | 0.60m | 0.60m |
| Size of fdn. raft : | 1.80m | 1.80m |
| Thickness of raft : | 0.40 m | |
| Original ground level: | 0.00 m | |
| Finished floor level : | 0.00 m | |
| U/S level of foundation : | -1.20 m | |
| U/S level Base Plate : | 0.60 m | |
| Unit weight of conc : | 24 kN/m$^3$ | |
| Unit weight of soil : | 20 kN/m$^3$ | |

**SYMBOLS AND NOTATIONS**

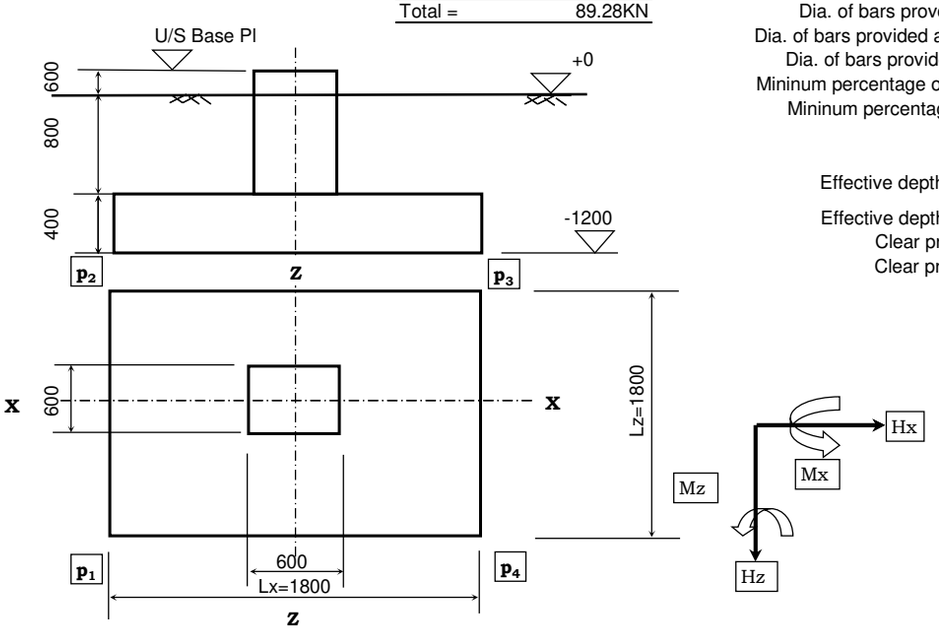
| | |
|---|---|
| Dead Load | D |
| Seismic Load | E |
| Wind Load | W |
| Live Load | L |
| Fluid Load | F |

**Calculation of Weight of Foundation :**

Weight of raft & Col.= (1.8*1.8*0.4*24)+(0.6*0.6*1.4*24) = 43.20KN

Weight of soil = ((1.8*1.8)-(0.6*0.6))*0.8*20 = 46.08KN

Total = 89.28KN




**Design Input:**

| | |
|---|---|
| Characteristic strength of concrete, f '$_c$= | 35 MPa |
| Yield stress of steel reinforcement, $f_y$= | 420 MPa |
| Clear cover for main reinforcement bars = | 50mm |
| Dia. of bars provd. along X direction at bottom = | 16mm |
| Dia. of bars provd. along X direction at top = | 10mm |
| Dia. of bars provided along Z direction at bottom= | 16mm |
| Dia. of bars provided along Z direction at top= | 10mm |
| Mininum percentage of reinforcement at bottom= | 0.18% |
| Mininum percentage of reinforcement at top= | 0.06% |

Effective depth of slab along X direction = 342mm = 0.342m

Effective depth of slab along Z direction = 326mm = 0.326m

Clear projection along Z direction = 0.6m

Clear projection along X direction = 0.6m

$H_X$=Force along X-X direction

Hz=Force along Z-Z direction

$M_X$=Moment about X-X direction

Mz=Moment about Z-Z direction


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| Sl. No. | Basic Load Case | V ( kN) | Hx (kN) | Hz (kN) | Mz (kNm) | Mx (kNm) |
|---|---|---|---|---|---|---|
| 1 | Dead Load (D) | 21.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2 | Seismic Load (E) | 0.00 | 3.50 | 3.50 | 0.00 | 0.00 |
| 3 | Wind Load (W) | 0.00 | 5.00 | 5.00 | 0.00 | 0.00 |
| 4 | Live Load (L) | 170.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 5 | Fluid Load (F) | 130.00 | 0.00 | 0.00 | 0.00 | 0.00 |

| **Combined Load Cases:** | | **Load at top of foundation** | | | | | **Load at bottom of foundation** | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Load Case | Load Combination | V (kN) | Hx (kN) | Mz (kN-m) | Hz (kN) | Mx (kN-m) | V (kN) | Hx (kN) | Mz (kN-m) | Hz (kN) | Mx (kN-m) |
| **Allowable Stress Design** | | | | | | | | | | | |
| 201 | D + F | 151.0 | 0.0 | 0.0 | 0.0 | 0.0 | 240.28 | 0.00 | 0.00 | 0.00 | 0.00 |
| 202 | D + F + L | 321.0 | 0.0 | 0.0 | 0.0 | 0.0 | 410.28 | 0.00 | 0.00 | 0.00 | 0.00 |
| 203 | D + F + W | 151.0 | 5.0 | 0.0 | 5.0 | 0.0 | 240.28 | 5.00 | 9.00 | 5.00 | 9.00 |
| 204 | 0.6 D + W | 12.6 | 5.0 | 0.0 | 5.0 | 0.0 | 101.88 | 5.00 | 9.00 | 5.00 | 9.00 |
| 205 | 0.6 D + 0.7 E | 12.6 | 2.5 | 0.0 | 2.5 | 0.0 | 66.17 | 2.45 | 4.41 | 2.45 | 4.41 |
| **Ultimate Conditions** | | | | | | | | | | | |
| 301 | 1.4 D + 1. 4F | 211.4 | 0.0 | 0.0 | 0.0 | 0.0 | 336.39 | 0.00 | 0.00 | 0.00 | 0.00 |
| 302 | 1.2 D + 1.2 F + 1.6 L | 453.2 | 0.0 | 0.0 | 0.0 | 0.0 | 578.19 | 0.00 | 0.00 | 0.00 | 0.00 |
| 303 | 1.2 D + 1.0 E + 0.5 L | 110.2 | 3.5 | 0.0 | 3.5 | 0.0 | 217.34 | 3.50 | 6.30 | 3.50 | 6.30 |
| 304 | 0.9 D + 1.0 E | 18.9 | 3.5 | 0.0 | 3.5 | 0.0 | 99.25 | 3.50 | 6.30 | 3.50 | 6.30 |
| 305 | 0.9 D + 1.6 W | 18.9 | 8.0 | 0.0 | 8.0 | 0.0 | 99.25 | 8.00 | 14.40 | 8.00 | 14.40 |
| 306 | 1.2 D + 1.6 W + 0.5 L | 110.2 | 8.0 | 0.0 | 8.0 | 0.0 | 217.34 | 8.00 | 14.40 | 8.00 | 14.40 |

**Check for overturning and sliding:**

| | Load Cases: | Load at b/o fdn | | | | | F.O.S. Mz | F.O.S. Mx | F.O.S. Hx | F.O.S. Hz |
|---|---|---|---|---|---|---|---|---|---|---|
| | | V (kN) | Hx (kN) | Mz (kNm) | Hz (kN) | Mx (kNm) | | | | |
| | | | | | | | | | | |
| 201 | D + F | 240.28 | 0.00 | 0.00 | 0.00 | 0.00 | - | - | - | - |
| 202 | D + F + L | 410.28 | 0.00 | 0.00 | 0.00 | 0.00 | - | - | - | - |
| 203 | D + F + W | 240.28 | 5.00 | 9.00 | 5.00 | 9.00 | 24.03 | 24.03 | 19.22 | 19.22 |
| 204 | 0.6 D + W | 101.88 | 5.00 | 9.00 | 5.00 | 9.00 | 10.19 | 10.19 | 8.15 | 8.15 |
| 205 | 0.6 D + 0.7 E | 66.17 | 2.45 | 4.41 | 2.45 | 4.41 | 13.50 | 13.50 | 10.80 | 10.80 |

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**Load Combination & Calculation of Base pressure**

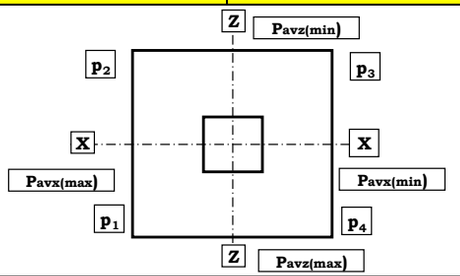
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m²

Net SBC of soil for Dead Load = 350 kN/m² For Seismic load case = 437.5 kN/m²

Gross SBC of soil for Dead Load = 374 kN/m² For Seismic load case = 461.5 kN/m²




| Load Case | Load Combination | Design Load/moments | | | Base Pressure (kN/m²) | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | V (kN) | Mz (kN-m) | Mx (kN-m) | V/ (Lx.Lz) | 6.Mz/ (Lx)².Lz | 6.Mx/ Lx.(Lz)² | $p_1$ ($p_{max}$) | $p_2$ | $p_4$ | $p_3$ ($p_{min}$) | ($p_{max}$-γh) | $p_{avx(max)}$ | $p_{avx(min)}$ | $p_{avz(max)}$ | $p_{avz(min)}$ | Remarks |
| 201 | D + F | 240.28 | 0.00 | 0.00 | 74.16 | 0.00 | 0.00 | 74.16 | 74.16 | 74.16 | 74.16 | 50.16 | 74.16 | 74.16 | 74.16 | 74.16 | O.K |
| 202 | D + F + L | 410.28 | 0.00 | 0.00 | 126.63 | 0.00 | 0.00 | 126.63 | 126.63 | 126.63 | 126.63 | 102.63 | 126.63 | 126.63 | 126.63 | 126.63 | O.K |
| 203 | D + F + W | 240.28 | 9.00 | 9.00 | 74.16 | 9.26 | 9.26 | 92.68 | 74.16 | 74.16 | 55.64 | 68.68 | 83.42 | 64.90 | 83.42 | 64.90 | O.K |
| 204 | 0.6 D + W | 101.88 | 9.00 | 9.00 | 31.44 | 9.26 | 9.26 | 49.96 | 31.44 | 31.44 | 12.93 | 25.96 | 40.70 | 22.19 | 40.70 | 22.19 | O.K |
| 205 | 0.6 D + 0.7 E | 66.17 | 4.41 | 4.41 | 20.42 | 4.54 | 4.54 | 29.50 | 20.42 | 20.42 | 11.35 | 5.50 | 24.96 | 15.89 | 24.96 | 15.89 | O.K |
| 301 | 1.4 D + 1. 4F | 336.39 | 0.00 | 0.00 | 103.82 | 0.00 | 0.00 | 103.82 | 103.82 | 103.82 | 103.82 | 79.82 | 103.82 | 103.82 | 103.82 | 103.82 | |
| 302 | 1.2 D + 1.2 F + 1.6 L | 578.19 | 0.00 | 0.00 | 178.45 | 0.00 | 0.00 | 178.45 | 178.45 | 178.45 | 178.45 | 154.45 | 178.45 | 178.45 | 178.45 | 178.45 | |
| 303 | 1.2 D + 1.0 E + 0.5 L | 217.34 | 6.30 | 6.30 | 67.08 | 6.48 | 6.48 | 80.04 | 67.08 | 67.08 | 54.12 | 56.04 | 73.56 | 60.60 | 73.56 | 60.60 | |
| 304 | 0.9 D + 1.0 E | 99.25 | 6.30 | 6.30 | 30.63 | 6.48 | 6.48 | 43.60 | 30.63 | 30.63 | 17.67 | 19.60 | 37.12 | 24.15 | 37.12 | 24.15 | |
| 305 | 0.9 D + 1.6 W | 99.25 | 14.40 | 14.40 | 30.63 | 14.81 | 14.81 | 60.26 | 30.63 | 30.63 | 1.00 | 36.26 | 45.45 | 15.82 | 45.45 | 15.82 | |
| 306 | 1.2 D + 1.6 W + 0.5 L | 217.34 | 14.40 | 14.40 | 67.08 | 14.81 | 14.81 | 96.71 | 67.08 | 67.08 | 37.45 | 72.71 | 81.90 | 52.27 | 81.90 | 52.27 | |
| Max : | | 578 | 14 | 14 | | | | | | | | | | | | | | |
| Min : | | 66 | 0 | 0 | | | | | | | | | | | | | | |

Max pressure at base of foundation without surcharge excluding weight of soil (Dead load cases) = 50.16 kN/m2

Max pressure at base of foundation including surcharge excluding weight of soil (Dead load cases) = 70.16 kN/m2

Max pressure at base of foundation without surcharge excluding weight of soil (Dead+Temperature+Seismic load cases) = 154.45 kN/m2

Max pressure at base of foundation including surcharge excluding weight of soil (Dead+ Temperature+Seismic load cases) = 174.45 kN/m2

Sample calculation of values of base pressure:

Load case : D + F + L

Maximum pressure, $p_1$ (pmax) = 126.63+0.00+0.00= 126.63 kN/m²

$p_2$= 126.63+0.00-0.00= 126.63 kN/m²

$p_3$= 126.63-(0.00+0.00)= 126.63 kN/m²

$p_4$= 126.63-0.00+0.00= 126.63 kN/m²

Maxm pressure excluding weight of soil ($p_{max}$-γh)= 126.63-1.2*20= 102.63 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (126.63+126.63)/2= 126.63 kN/m2

Minimum average Pressure along X, $p_{avx(minx)}$ = (126.63+126.63)/2= 126.63 kN/m2

Maximum average Pressure along Z, $p_{avz(max)}$ = (126.63+126.63)/2= 126.63 kN/m2

Minimum average Pressure along Z, $p_{avz(minx)}$ = (126.63+126.63)/2= 126.63 kN/m2


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**Calculation of Bending Moment and Shear Force in X direction :**

$p_{1x}$ =pressure at the face of column at section 1x-1x in kN/m2.

$p_{2x}$ =pressure at the face of column at section 2x-2x in kN/m2.

$p_{3x}$ =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m2.

$BM_{1x}$ =Moment at the face of column at section 1x-1x in kN-m/m.

$BM_{2x}$ =Moment at the face of column 2x-2x in kN-m/m.

$SF_{3x}$ = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.326)+(0.6+2*0.5*0.326))*2

= 3.704 m

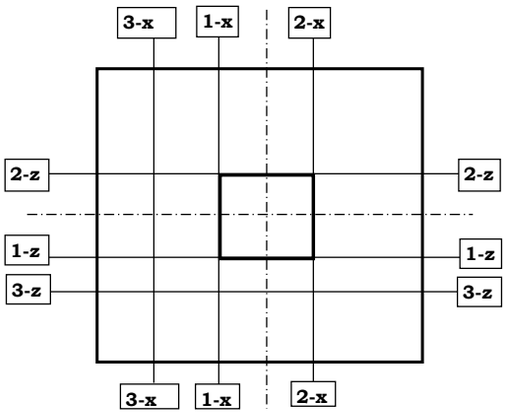
Punching Area= ((0.6+2*0.5*0.326)*(0.6+2*0.5*0.326))

= 0.86 m2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+0.8*20)

= 25.60 kN/m2

Area of footing, Af= 3.24 m2




| Load Case | Load Combination | $p_{1x}$ kN/m2 | $p_{2x}$ kN/m2 | $p_{3x}$ kN/m2 | B.$M_{1x}$ kNm | B.$M_{2x}$ kNm | S.$F_{3x}$ kN | S.F(punching) kN |
|---|---|---|---|---|---|---|---|---|
| Ultimate Conditions | | | | | | | | |
| 301 | 1.4 D + 1. 4F | 103.82 | 103.82 | 103.82 | 14.08 | 14.08 | 20.18 | 186.17 |
| 302 | 1.2 D + 1.2 F + 1.6 L | 178.45 | 178.45 | 178.45 | 27.51 | 27.51 | 39.44 | 363.79 |
| 303 | 1.2 D + 1.0 E + 0.5 L | 69.24 | 64.92 | 71.70 | 8.37 | 6.56 | 12.13 | 98.72 |
| 304 | 0.9 D + 1.0 E | 32.80 | 28.47 | 35.26 | 1.81 | 0.00 | 2.73 | 11.98 |
| 305 | 0.9 D + 1.6 W | 35.57 | 25.70 | 41.20 | 2.98 | -1.17 | 4.57 | 11.98 |
| 306 | 1.2 D + 1.6 W + 0.5 L | 72.02 | 62.15 | 77.65 | 9.54 | 5.39 | 13.98 | 98.72 |

Sample calculation of BM & SF

Load case : 301

$p_{1x}$ = 103.82-(103.82-103.82)*0.6/1.8 = 103.82 $kN/m^2$

$p_{2x}$ = 103.82+(103.82-103.82)*0.6/1.8 = 103.82 $kN/m^2$

$p_{3x}$ = 103.82-(103.82-103.82)*(0.6-0.342)/1.8 = 103.82 $kN/m^2$

Overhang = 0.6 m

Effective depth = 0.342 m


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$BM_{1x}$ = (103.82-25.6)*0.6*0.6/2+0.5*(103.82-103.82)*0.6*0.6*0.67 = 14.08 kN-m/m

$BM_{2x}$ = (103.82-25.6)*0.6*0.6/2-0.5*(-103.82+103.82)*0.6*0.6*0.67 = 14.08 kN-m/m

$SF_{3x}$ = (103.82-25.6)*(0.6-0.342)+0.5*(103.82-103.82)*(0.6-0.342) = 20.18 kN/m

$S.F_{(punching)}$ = (3.24-0.86)*(103.82-25.6) = 186.17 kN

| Location | Mux (kN-m) | $\phi$ Kn = $M_{u1}/bd^2$ ($N/mm^2$) | Kn ($N/mm^2$) | ρ= | Ast ($mm^2$) | Minm. Ast ($mm^2$) | Bar Dia | Spacing reqd.(mm) | Maxm spacing (mm) | Spacing provd. (mm) | Ast ($mm^2$) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| At bottom | 27.51 | 0.24 | 0.31 | 0.001 | 256.75 | 615.60 | 16ϕT | 326.61 | 300 | 200 | 1005.31 |
| At top | -1.17 | -0.01 | 0.01 | 0.000 | 10.84 | 205.20 | 10ϕT | 382.75 | 300 | 200 | 392.70 |

**Check against one-way shear force at a distance d from face of column:**

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 39.44 kN/m

Shear stress , $\tau_v$= (39.44*1000)/(1000*342)

= 0.12 N/mm2 **O.K.**

**Check against Two-way shear force at a distance d/2 from face of column:**

Design punching shear at a distance d/2 from face of column, Vu= 363.79 kN

Strength of Concrete in shear Vc1 = 3747.15 kN Table 22.6.5.2_ (a)

Strength of Concrete in shear Vc2 = 2402.33 kN Table 22.6.5.2_ (b)

Strength of Concrete in shear Vc3 = 2498.10 kN Table 22.6.5.2_ (c)

Design Vc= 2402.33 kN

Allowable shear stress, $(\phi\ Vc)/bd$ = 1.42 N/mm2

Design punching shear stress, τ = (363.79*1000)/(3704*342)

= 0.29 N/mm2 **No shear reinforcement reqd.**

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**Calculation of Bending Moment and Shear Force in Z direction :**

$p_{1z}$ =pressure at the face of column at section 1z-1z in kN/m2.
$p_{2z}$ =pressure at the face of column at section 2z-2z in kN/m2.
$p_{3z}$ =pressure at a distance d from the face of column at section 3z-3z in kN/m2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

| Load Case | Load Combination | $p_{1z}$ kN/m2 | $p_{2z}$ kN/m2 | $p_{3z}$ kN/m2 | $B.M_{1z}$ kNm | $B.M_{2z}$ kNm | $S.F_{3z}$ kN |
|---|---|---|---|---|---|---|---|
| Ultimate Conditions | | | | | | | |
| 301 | 1.4 D + 1. 4F | 103.82 | 103.82 | 103.82 | 14.08 | 14.08 | 21.43 |
| 302 | 1.2 D + 1.2 F + 1.6 L | 178.45 | 178.45 | 178.45 | 27.51 | 27.51 | 41.88 |
| 303 | 1.2 D + 1.0 E + 0.5 L | 69.24 | 64.92 | 71.59 | 8.38 | 6.56 | 12.87 |
| 304 | 0.9 D + 1.0 E | 32.80 | 28.47 | 35.15 | 1.82 | 0.00 | 2.89 |
| 305 | 0.9 D + 1.6 W | 35.57 | 25.70 | 40.94 | 2.99 | -1.17 | 4.82 |
| 306 | 1.2 D + 1.6 W + 0.5 L | 72.02 | 62.15 | 77.39 | 9.55 | 5.39 | 14.81 |

Sample calculation of BM & SF

Load case : 301

$p_{1z}$ = 103.82-(103.82-103.82)*0.6/1.8= 103.82 kN/m2

$p_{2z}$ = 103.82+(103.82-103.82)*0.6/1.8= 103.82 kN/m2 Overhang = 0.6 m

$p_{3z}$ = 103.82-(103.82-103.82)*(0.6-0.326)/1.8= 103.82 kN/m2 Effective depth = 0.326 m

$BM_{1z}$ = (103.82-25.6)*0.6*0.6/2+0.5*(103.82-103.82)*0.6*0.6*0.67 = 14.08 kN-m/m

$BM_{2z}$ = (103.82-25.6)*0.6*0.6/2-0.5*(-103.82+103.82)*0.6*0.6*0.67 = 14.08 kN-m/m

$SF_{3z}$ = (103.82-25.6)*(0.6-0.326)+0.5*(103.82-103.82)*(0.6-0.326) = 21.43 kN/m


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| Location | Mux (kN-m) | $\phi$ Kn = $M_{u1}/bd^2$ ($N/mm^2$) | Kn (N/mm2) | ρ= | Ast (mm2) | Minm. Ast (mm2) | Bar Dia | Spacing reqd.(mm) | Maxm spacing (mm) | Spacing provd. (mm) | Ast (mm2) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| At bottom | 27.51 | 0.259 | 0.345 | 0.0008 | 282.72 | 586.80 | 16ϕT | 342.64 | 300 | 200 | 1005.3 |
| At top | -1.17 | -0.011 | -0.015 | 0.0000 | -0.11 | 195.60 | 10ϕT | 401.53 | 300 | 200 | 392.7 |

**Check against one-way shear force at a distance d from face of column:**

Allowable shear stress, ($\phi$ Vc) = 0.74 $N/mm^2$

Shear at a distance 'd' from face of column, $V_u$= 41.88 kN/m

Shear stress , $\tau_v$= (41.88*1000)/(1000*326)

= 0.13 $N/mm^2$ **O.K.**

**SUMMARY OF REINFORCEMENT IN RAFT:**

| | | | | |
|---|---|---|---|---|
| Along X Direction : | | | | |
| | At Bottom | 16ϕT | 200 | c/c |
| | At Top | 10ϕT | 200 | c/c |
| Along Z Direction : | | | | |
| | At Bottom | 16ϕT | 200 | c/c |
| | At Top | 10ϕT | 200 | c/c |


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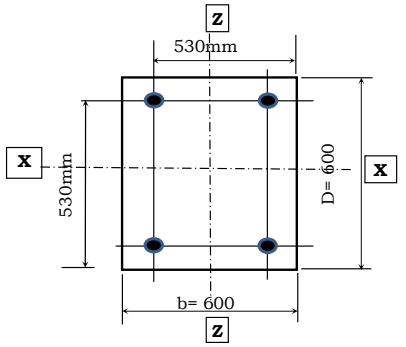
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**DESIGN OF COLUMN FOR 3610-TNK-001**

**Input:**

| | |
|---|---|
| Clear cover to reinf.= | 50mm |
| Dia. of main reinf.= | 20mm |
| Dia. of link= | 10mm |
| Effective cover = | 70mm |
| Grade of Concrete ($f_c'$) = | M35 |
| Grade of Steel ($f_y$) = | Fy420 |
| Grade of Reinf(links).= | Fy420 |
| Min. Steel Ratio | 0.50% |
| Max. Steel Ratio | 4.00% |

Table 20.6.1.3.1 of SBC-304




For Compression-controlled sections

Strength reduction factor $\phi$ = 0.65 Table 21.2.2 of SBC-304

| | |
|---|---|
| Unsupproted length of column about X ($l_{ux}$) = | 1400mm |
| Unsupproted length of column about Z ($l_{uz}$) = | 1400mm |

**For Columns not braced against sidesway,**

Where, Effective length factor K= 2

Slenderness ratio, $K l_{ux} / r_x$ = 15.56 < 22 SECTION 6.2.5a Radius of gyration $r_x$= 180 mm

$K l_{uz} / r_z$ = 15.56 < 22 SECTION 6.2.5a Radius of gyration $r_z$= 180 mm

| Sl. No. | Basic Load Case | V ( kN) | Hx (kN) | Hz (kN) | Mz (kNm) | Mx (kNm) |
|---|---|---|---|---|---|---|
| 1 | Dead Load (D) | 33.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 2 | Seismic Load (E) | 0.00 | 3.50 | 3.50 | 0.00 | 0.00 |
| 3 | Wind Load (W) | 0.00 | 5.00 | 5.00 | 0.00 | 0.00 |
| 4 | Live Load (L) | 170.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 5 | Fluid Load (F) | 130.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| | | | $\gamma_x$ = | 0.80 | | |
| | | | $\gamma_z$ = | 0.80 | | |


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| Load Case | Load Combination | Factored Load/Moment | | | Nominal Load/Moment | | | Kn = Pn/f'cbD (N/mm2) |
|---|---|---|---|---|---|---|---|---|
| | | Pu (T) | Muz (Tm) | Mux (Tm) | Pn (T) | Mnz (Tm) | Mnx (Tm) | |
| 301 | 1.4 D + 1.4 F | 22.82 | 0.00 | 0.00 | 35.11 | 0.00 | 0.00 | 0.03 |
| 302 | 1.2 D + 1.2 F + 1.6 L | 46.76 | 0.00 | 0.00 | 71.94 | 0.00 | 0.00 | 0.06 |
| 303 | 1.2 D + 1.0 E + 0.5 L | 12.46 | 0.49 | 0.49 | 19.17 | 0.75 | 0.75 | 0.02 |
| 304 | 0.9 D + 1.0 E | 2.97 | 0.49 | 0.49 | 4.57 | 0.75 | 0.75 | 0.00 |
| 305 | 0.9 D + 1.6 W | 2.97 | 1.12 | 1.12 | 4.57 | 1.72 | 1.72 | 0.00 |
| 306 | 1.2 D + 1.6 W + 0.5 L | 12.46 | 1.12 | 1.12 | 19.17 | 1.72 | 1.72 | 0.02 |

**Reinforcement:** Provide,

| | | | |
|---|---|---|---|
| 4-Nos | 20 mm Dia | 1257.14 Sq mm | (At Corner) |
| 4-Nos | 20 mm Dia | 1257.14 Sq mm | (Along Z axis) |
| 4-Nos | 20 mm Dia | 1257.14 Sq mm | (Along Y axis) |

Asc pvd. = 3771.43 Sq mm

Total Asc = 1.0 %

$\beta$ = $P_n / P_0$ (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1218.18 T SECTION 22.4.2.2

| Load Case | From Chart of SP17(M)-R35-420.8 | | $M_{noz}$ | $M_{nox}$ | $\beta$ | $\alpha_n = \log(0.5)/\log(\beta)$ | Interaction ratio | Remarks |
|---|---|---|---|---|---|---|---|---|
| | $R_{nz}=M_{noz}/f'_c b^2 D$ | $R_{nx}=M_{nox}/f'_c b^2 D$ | | | | | | |
| 301 | 0.05 | 0.05 | 37.8 | 37.8 | 0.55 | 1.16 | 0.00 | O.K |
| 302 | 0.05 | 0.05 | 37.8 | 37.8 | 0.55 | 1.16 | 0.00 | O.K |
| 303 | 0.05 | 0.05 | 37.8 | 37.8 | 0.55 | 1.16 | 0.02 | O.K |
| 304 | 0.05 | 0.05 | 37.8 | 37.8 | 0.55 | 1.16 | 0.02 | O.K |
| 305 | 0.05 | 0.05 | 37.8 | 37.8 | 0.55 | 1.16 | 0.06 | O.K |
| 306 | 0.05 | 0.05 | 37.8 | 37.8 | 0.55 | 1.16 | 0.06 | O.K |

Links, 10mm dia @ 300 C/C


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## DESIGN CALCULATION OF FOUNDARION FOR 3650-COL-001 & 002

| Size of Columns at node 7787,7973,7971 & 7968 | Along-Z 0.50m | Along-X 0.50m |
|---|---|---|
| Size of fdn. raft: | 2.50m | 2.50m |
| Thickness of raft: | 0.40 m | |
| Original Gr. Level: | 0.00 m | |
| Finished Gr. Level: | 0.00 m | |
| U/S of B/Pl Above GL: | 0.55 m | |
| U/S level of fdn: | -1.20 m | |
| Unit weight of conc: | 25 kN/m$^3$ | |
| Unit weight of soil: | 20 kN/m$^3$ | |

**Calculation of Weight of Foundation :**

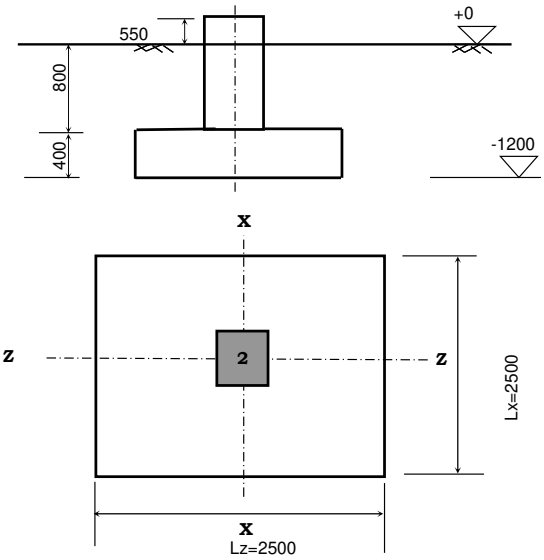
Weight of raft = (2.5*2.5*0.4*25) = 62.50KN

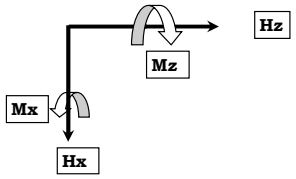
Weight of soil = ((2.5*2.5)-(0.5*0.5))*0.8*20 = 96.00KN

Total = 158.50KN

**Design Input:**

| | |
|---|---|
| Characteristic strength of concrete, f '$_c$= | 35 MPa |
| Yield stress of steel reinforcement, $f_y$= | 420 MPa |
| Clear cover for main reinforcement bars = | 50mm |




**$H_x$=Force along X-X direction**

**Hz=Force along Z-Z direction**

**$M_x$=Moment about X-X direction**

**Mz=Moment about Z-Z direction**


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**Check for overturning and sliding**

| Load at bottom of fooring | Node 7787 | | | | | Node 7971 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| L/C No. | V (kN) | Hx (kN) | Mz (kN-m) | Hz (kN) | Mx (kN-m) | V (kN) | Hx (kN) | Mz (kN-m) | Hz (kN) | Mx (kN-m) |
| 201 D + F | 20.94 | 0.00 | 0.00 | 0.00 | 0.00 | 20.94 | 0.00 | 0.00 | 0.00 | 0.00 |
| 202 D + F + L | 70.94 | 0.00 | 0.00 | 0.00 | 0.00 | 70.94 | 0.00 | 0.00 | 0.00 | 0.00 |
| 203 D + F + W | 120.94 | -8.00 | 14.00 | -8.00 | -14.00 | 120.94 | -8.00 | 14.00 | -8.00 | -14.00 |
| 204 0.6D + W | 112.56 | -8.00 | 14.00 | -8.00 | -14.00 | 112.56 | -8.00 | 14.00 | -8.00 | -14.00 |
| 205 0.6D + 0.7E | 52.46 | -2.80 | 4.90 | -2.80 | -4.90 | 52.46 | -2.80 | 4.90 | -2.80 | -4.90 |
| 206 D + 0.75W + 0.75L | 133.44 | -6.00 | 10.50 | -6.00 | -10.50 | 133.44 | -6.00 | 10.50 | -6.00 | -10.50 |

| Load at bottom of fooring | Node 7973 | | | | | Node 7968 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| L/C No. | V (kN) | Hx (kN) | Mz (kN-m) | Hz (kN) | Mx (kN-m) | V (kN) | Hx (kN) | Mz (kN-m) | Hz (kN) | Mx (kN-m) |
| 201 D + F | 20.94 | 0.00 | 0.00 | 0.00 | 0.00 | 20.938 | 0 | 0 | 0 | 0 |
| 202 D + F + L | 70.94 | 0.00 | 0.00 | 0.00 | 0.00 | 70.938 | 0 | 0 | 0 | 0 |
| 203 D + F + W | 120.94 | -8.00 | 14.00 | -8.00 | -14.00 | 120.938 | -8 | 14 | -8 | -14 |
| 204 0.6D + W | 112.56 | -8.00 | 14.00 | -8.00 | -14.00 | 112.563 | -8 | 14 | -8 | -14 |
| 205 0.6D + 0.7E | 52.46 | -2.80 | 4.90 | -2.80 | -4.90 | 52.463 | -2.8 | 4.9 | -2.8 | -4.9 |
| 206 D + 0.75W + 0.75L | 133.44 | -6.00 | 10.50 | -6.00 | -10.50 | 133.438 | -6 | 10.5 | -6 | -10.5 |

**Check for overturning and sliding:**

| | Resultant Load at bottom of footing | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| L/C No. | V (kN) | Hz (kN) | Mx (kN-m) | Hx (kN) | Mz (kN-m) | Overturning | | Sliding | |
| | | | | | | F.O.S. Mz | F.O.S. Mx | F.O.S. Hx | F.O.S. Hz |
| 201 D + F | 242.3 | 0.00 | 0.0 | 0.0 | 0.0 | NA | NA | NA | NA |
| 202 D + F + L | 442.3 | 0.00 | 0.0 | 0.0 | 0.0 | NA | NA | NA | NA |
| 203 D + F + W | 642.3 | -32.00 | -56.0 | -32.0 | 56.0 | 14.3 | 14.3 | 8.0 | 8.0 |
| 204 0.6D + W | 608.8 | -32.00 | -56.0 | -32.0 | 56.0 | 13.6 | 13.6 | 7.6 | 7.6 |
| 205 0.6D + 0.7E | 368.4 | -11.20 | -19.6 | -11.2 | 19.6 | 23.5 | 23.5 | 13.2 | 13.2 |
| 206 D + 0.75W + 0.75L | 692.3 | -24.00 | -42.0 | -24.0 | 42.0 | 20.6 | 20.6 | 11.5 | 11.5 |


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Base pressure check

| | Node | L/C | Horizontal Fx kN/m2 | Vertical Fy kN/m2 | Horizontal Fz kN/m2 |
|---|---|---|---|---|---|
| Max Px | 7787 | 201 | 0 | 59.89 | 0 |
| Min Px | 7787 | 201 | 0 | 59.89 | 0 |
| Max Py | 8041 | 203 | 0 | 158.40 | 0 |
| Min Py | 8051 | 205 | 0 | 44.27 | 0 |
| Max Pz | 7787 | 201 | 0 | 59.89 | 0 |
| Min Pz | 7787 | 201 | 0 | 59.89 | 0 |

Gross Maximum Base pressure, $p_{max}$ = 158.4 $KN/m^2$

Net Allowable Safe Bearing Capacity = 350 $KN/m^2$
Depth of soil above footing = 0.8 m
Unit weight of soil = 20 $KN/m^3$

Gross Safe Bearing Capacity = 366 $KN/m^2$
> 158.4 $KN/m^2$
**Hence OK**


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Design of foundation raft

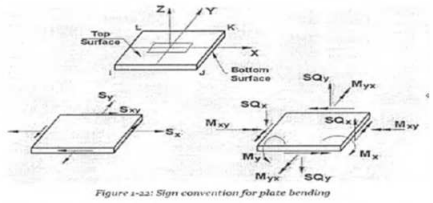
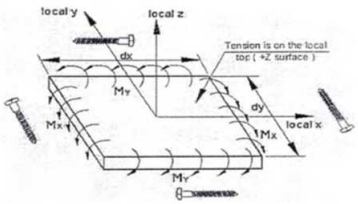


Figure 1-22: Sign convention for plate bending




**Sign convention :**
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

**Moment :**
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

**Shear force :**
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

**Axial force :**
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

**Reinforcement in Foundation:**
PLATE CENTER STRESS : FDN_MOMENT At the face of the column

| | | Shear | | Membrane | | | Bending Moment | | | M + I $M_{xy}$ I | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Plate | Qx | Qy | Sx | Sy | Sxy | Mx | My | Mxy | Mx | My | L/C |
| | | ($N/mm^2$) | ($N/mm^2$) | ($N/mm^2$) | ($N/mm^2$) | ($N/mm^2$) | (kN-m/m) | (kN-m/m) | (kN-m/m) | (kN-m/m) | (kN-m/m) | |
| Max Qx | 297 | 0.305 | -0.3 | 0 | 0 | 0 | 13.816 | 50.945 | 0.632 | 14.448 | 51.577 | 106:1.2D + 1.6W + 0.5L |
| Min Qx | 302 | -0.492 | -0.446 | 0 | 0 | 0 | 36.466 | 59.233 | 12.033 | 48.499 | 71.266 | 106:1.2D + 1.6W + 0.5L |
| Max Qy | 282 | -0.3 | 0.306 | 0 | 0 | 0 | 50.973 | 13.804 | 0.622 | 51.595 | 14.426 | 106:1.2D + 1.6W + 0.5L |
| Min Qy | 312 | -0.446 | -0.492 | 0 | 0 | 0 | 59.18 | 36.476 | 12.026 | 71.206 | 48.502 | 106:1.2D + 1.6W + 0.5L |
| Max Sx | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Min Sx | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Max Sy | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Min Sy | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Max Sxy | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Min Sxy | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Max Mx | 312 | -0.446 | -0.492 | 0 | 0 | 0 | 59.18 | 36.476 | 12.026 | 71.206 | 48.502 | 106:1.2D + 1.6W + 0.5L |
| Min Mx | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Max My | 302 | -0.492 | -0.446 | 0 | 0 | 0 | 36.466 | 59.233 | 12.033 | 48.499 | 71.266 | 106:1.2D + 1.6W + 0.5L |
| Min My | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Max Mxy | 302 | -0.443 | -0.402 | 0 | 0 | 0 | 33.123 | 52.781 | 12.123 | 45.246 | 64.904 | 105:0.9D + 1.6W |
| Min Mxy | 305 | 0.093 | 0.005 | 0 | 0 | 0 | 29.925 | 23 | -14.768 | 44.693 | 37.768 | 106:1.2D + 1.6W + 0.5L |

PLATE CENTER STRESS : FDN_SHEAR At a distance "d" from the face of the column

| | | Shear | | Membrane | | | Bending Moment | | | M + I $M_{xy}$ I | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Plate | Qx | Qy | Sx | Sy | Sxy | Mx | My | Mxy | Mx | My | L/C |
| | | ($N/mm^2$) | ($N/mm^2$) | ($N/mm^2$) | ($N/mm^2$) | ($N/mm^2$) | (kN-m/m) | (kN-m/m) | (kN-m/m) | (kN-m/m) | (kN-m/m) | |
| Max Qx | 286 | 0.141 | 0.141 | 0 | 0 | 0 | 21.805 | 21.792 | 1.88 | 23.685 | 23.672 | 106:1.2D + 1.6W + 0.5L |
| Min Qx | 313 | -0.186 | -0.078 | 0 | 0 | 0 | 26.084 | 16.418 | 1.528 | 27.612 | 17.946 | 106:1.2D + 1.6W + 0.5L |
| Max Qy | 286 | 0.141 | 0.141 | 0 | 0 | 0 | 21.805 | 21.792 | 1.88 | 23.685 | 23.672 | 106:1.2D + 1.6W + 0.5L |
| Min Qy | 308 | -0.077 | -0.186 | 0 | 0 | 0 | 16.378 | 26.126 | 1.529 | 17.907 | 27.655 | 106:1.2D + 1.6W + 0.5L |
| Max Sx | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Min Sx | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Max Sy | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Min Sy | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Max Sxy | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Min Sxy | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Max Mx | 338 | 0.055 | -0.095 | 0 | 0 | 0 | 40.888 | 3.155 | 1.518 | 42.406 | 4.673 | 106:1.2D + 1.6W + 0.5L |
| Min Mx | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Max My | 317 | -0.095 | 0.056 | 0 | 0 | 0 | 3.164 | 40.907 | 1.509 | 4.673 | 42.416 | 106:1.2D + 1.6W + 0.5L |
| Min My | 342 | -0.003 | -0.003 | 0 | 0 | 0 | 0.198 | 0.197 | -0.012 | 0.21 | 0.209 | 101:1.4D + 1.4F |
| Max Mxy | 307 | -0.137 | -0.137 | 0 | 0 | 0 | 31.922 | 31.97 | 6.491 | 38.413 | 38.461 | 106:1.2D + 1.6W + 0.5L |
| Min Mxy | 329 | 0.031 | 0.013 | 0 | 0 | 0 | 17.825 | 1.048 | -4.644 | 22.469 | 5.692 | 106:1.2D + 1.6W + 0.5L |


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**Material Property :**

| | | | |
|---|---|---|---|
| Grade of concrete = | | fc'35 | |
| Grade of reinf. = | | fy420 | |
| Clear cover = | bottom face | 50 | mm |
| | top face | 50 | mm |
| Thicknes of slab = | | 400 | mm |

**Design Constants :**

| | |
|---|---|
| φ for Moment = | 0.9 |
| φ for shear = | 0.75 |
| $\beta_1$ | 0.8 |
| $\varepsilon_t$ limit (for tension controlled section) = | 0.005 |
| Concrete compressive strain, $\varepsilon_{cu}$ = | 0.003 |

**Reinforcement Along X:**

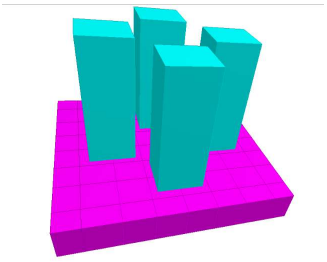
| Force Values | | | Depth | | Reinforcement Calculation | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Load condition | Factored Mt, Mx | Tension face due to bending | clear cover | d | $M_u$ | Kn = Mu1/bd2 / ϕ | ρ | $A_{s\ required}$ | $A_{s,min}$ | Dia of bar | $S_{required}$ | $S_{provided}$ | $A_{s,\ provided}$ | a | c = a/$\beta_1$ | $\varepsilon_t$ = $\varepsilon_{cu}$/c *(d-c) | Remarks |
| | KN-m/m | | mm | mm | KN-m/m | (N/mm2 | mm | $mm^2$ | $mm^2$ | mm | mm c/c | mm c/c | $mm^2$ | mm | mm | | |
| 106:1.2D | 71.21 | bottom face | 50 | 344 | 71.206 | 0.80 | 0.0019 | 666.2 | 619.2 | 12 | 169.82 | 150 | 754.3 | 10.6 | 13.29 | 0.075 | tension control |
| 101:1.4D | 0.21 | bottom face | 50 | 344 | 0.21 | 0.00 | 0.0000 | 1.9 | | 12 | 182.72 | 150 | 754.3 | 10.6 | 13.29 | 0.075 | tension control |
| 105:0.9D | 45.25 | bottom face | 50 | 344 | 45.246 | 0.51 | 0.0012 | 421.2 | | 12 | 182.72 | 150 | 754.3 | 10.6 | 13.29 | 0.075 | tension control |
| 106:1.2D | 44.69 | bottom face | 50 | 344 | 44.693 | 0.50 | 0.0012 | 416.0 | | 12 | 182.72 | 150 | 754.3 | 10.6 | 13.29 | 0.075 | tension control |

| Shear Check | | Qx | FOS | Vu | Mu | $\rho_{w,pro}$ | bd | $V_{ca}$ | $V_{cb}$ | $V_{cc}$ | $V_{c,design}$ | $\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$ | | Remarks | $\varphi V_c$ | Remarks | Remarks :Min shear reinf |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Mpa | | KN | KN-m/m | | $m^2$ | KN | KN | KN | KN | KN | | | KN | | |
| Load condition | Max Qx | 0.141 | 1 | 56.4 | 23.685 | 0.002 | 0.344 | 336.125 | 338.444 | 590.188 | 336.13 | 1259 | > 56.4 | ok | 252.1 | φVc>Vu | not required |
| | Min Qx | 0.186 | 1 | 74.4 | 27.612 | | | 337.507 | | | 337.51 | 1261 | > 74.4 | ok | 253.1 | φVc>Vu | not required |

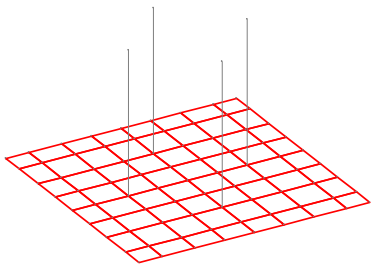
**Reinforcement Along Z:**

| Force Values | | | Depth | | Reinforcement Calculation | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Load condition | Factored Mt, My | Tension face due to bending | clear cover | d | $M_u$ | Kn = Mu1/bd2 / ϕ | ρ | $A_{s\ required}$ | $A_{s,min}$ | Dia of bar | $S_{required}$ | $S_{provided}$ | $A_{s,\ provided}$ | a | c = a/$\beta_1$ | $\varepsilon_t$ = $\varepsilon_{cu}$/c *(d-c) | Remarks |
| | KN-m/m | | mm | mm | KN-m/m | (N/mm2 | mm | $mm^2$ | $mm^2$ | mm | mm c/c | mm c/c | $mm^2$ | mm | mm | | |
| 106:1.2D | 71.27 | bottom face | 50 | 344 | 71.266 | 0.80 | 0.0019 | 666.8 | 619.2 | 12 | 169.68 | 150 | 754.3 | 10.6 | 13.29 | 0.075 | tension control |
| 101:1.4D | 0.21 | bottom face | 50 | 344 | 0.209 | 0.00 | 0.0000 | 1.9 | | 12 | 182.72 | 150 | 754.3 | 10.6 | 13.29 | 0.075 | tension control |
| 105:0.9D | 64.90 | bottom face | 50 | 344 | 64.904 | 0.73 | 0.0018 | 606.5 | | 12 | 182.72 | 150 | 754.3 | 10.6 | 13.29 | 0.075 | tension control |
| 106:1.2D | 37.77 | bottom face | 50 | 344 | 37.768 | 0.43 | 0.0010 | 351.1 | | 12 | 182.72 | 150 | 754.3 | 10.6 | 13.29 | 0.075 | tension control |

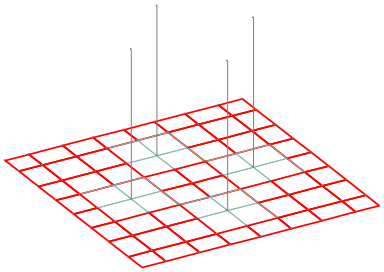
| Shear Check | | Qx | FOS | Vu | Mu | $\rho_{w,pro}$ | bd | $V_{ca}$ | $V_{cb}$ | $V_{cc}$ | $V_{c,design}$ | $\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$ | | Remarks | $\varphi V_c$ | Remarks | Remarks :Min shear reinf |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Mpa | | KN | T-m/m | | $m^2$ | KN | KN | KN | KN | KN | | | KN | | |
| Load condition | Max Qy | 0.141 | 1 | 56.4 | 23.672 | 0.002 | 0.344 | 336.131 | 338.444 | fc'590.2 | 336.13 | 353 | > 56.4 | ok | 252.1 | φVc>Vu | not required |
| | Min Qy | 0.186 | 1 | 74.4 | 27.655 | | | 337.488 | | | 337.49 | 354 | > 74.4 | ok | 253.1 | φVc>Vu | not required |



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

**SUMMARY OF REINFORCEMENT IN RAFT:**

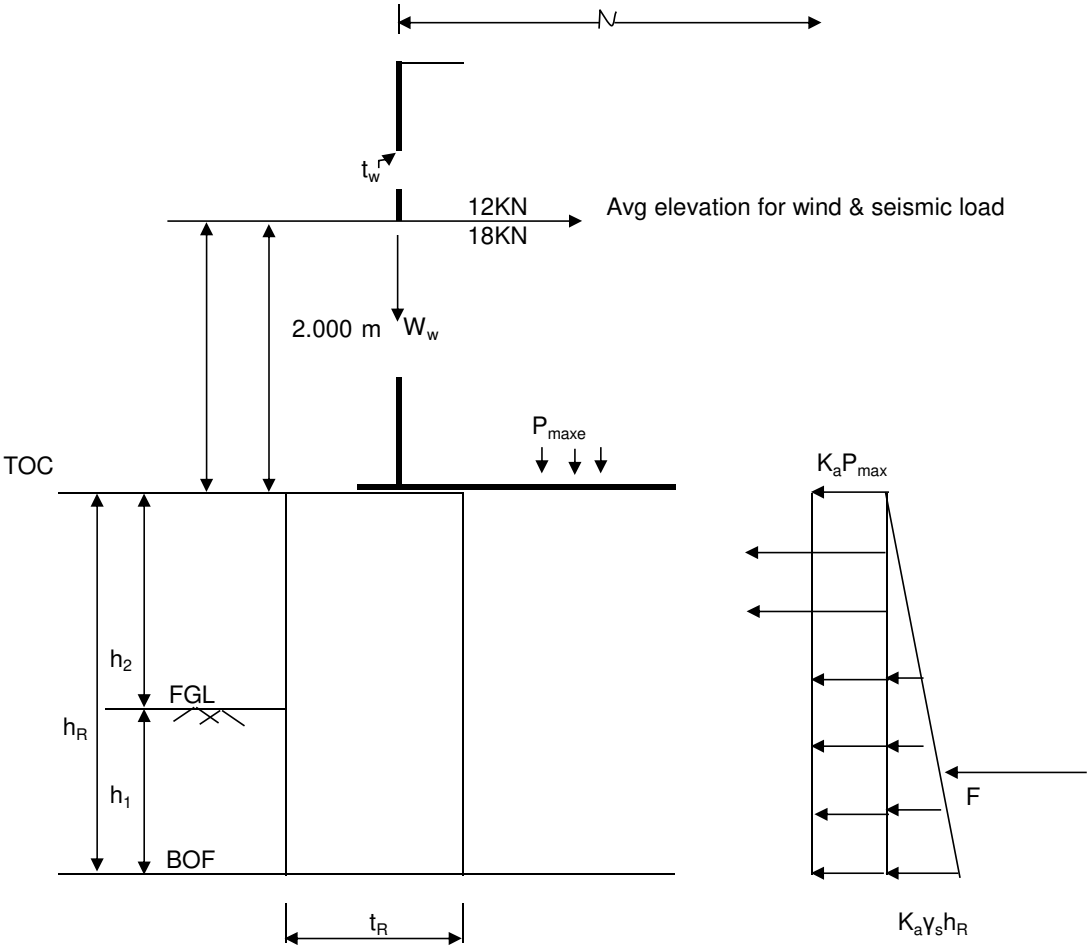
| | | | |
|---|---|---|---|
| Along X Direction : | | | |
| At Bottom | 12ϕT | 150 | c/c |
| At Top | 10ϕT | 150 | c/c |
| Along Z Direction : | | | |
| At Bottom | 12ϕT | 150 | c/c |
| At Top | 10ϕT | 200 | c/c |


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Design Calculation of Foundation for 3650-TNK-001 & 3650-TNK-003




Typical Section Through Tank

Pressure on ring wall


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Tank Data :

| | | |
|---|---|---|
| Diameter of tank ,D = | 2.5 | m |
| Self weight of tank = | 25 | KN |
| Self weight of tank per m of circumference,$W_w$ = | 3.18 | KN/m |
| Maximum weight of Fluid = | 220 | KN |
| Pressure at base of tank ,$P_{max}$ | 44.8 | $KN/m^2$ |
| Design height of liquid, $h_L$ = | 3.5 | m |

Foundation Material :

| | | |
|---|---|---|
| Concrete, $f_c'$ = | 35 | Mpa |
| Reinforcement, $f_y$ = | 420 | Mpa |
| Unit weight of concrete,$\gamma_c$ = | 24 | $Kn/m^3$ |

Soil Data :

| | | |
|---|---|---|
| Unit weight of soil,$\gamma_s$ = | 20 | $Kn/m^3$ |
| Allowable bearing pressure, $q_{all}$= | 375 | $Kn/m^2$ |
| Coefficient of active earth pressure,$K_a$ = | 0.33 | |

Design Data of Ringwall :

| | | |
|---|---|---|
| Ringwall depth, $h_R = h_1 + h_2$ = | 1.8 | m |
| FGL to Bottom of foundation ,$h_1$ = | 1.2 | m |
| FGL to Top of concrete ,$h_2$ = | 0.6 | m |
| Ringwall thickness, $t_R$ = | 450 | mm |
| Inner dia of ringwall,$D_i$ = | 2.05 | m |
| Outer dia of ringwall,$D_o$ = | 2.95 | m |
| $S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ = | 1.93 | $m^3$ |


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Check of soil bearing pressure (Max Fluid Load): For wind Load

Wind force = 18 kn acting at a height of 3.800 m from BOF

$M_{OT}$ = Overturning moment = 68.4 KN-m

$M_{OT} / S_{xx}$ = 35.4 $KN/m^2$

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 72.7 $KN/m^2$

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 108 $KN/m^2$ < 375.0 $Kn/m^2$ OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 37.3 $KN/m^2$

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = W_w/t_R + h_R\gamma_c$ = 50.3 $KN/m^2$

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 85.7 $KN/m^2$

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 14.88 $KN/m^2$ no -ve pressure

Check for overturning (Empty Tank): For wind Load

$M_{OT}$ = Overturning moment = 68.4 KN-m

$M_{OR}$ = Restoring moment = (25+PI()*2.5*0.45*1.8*25)*2.5/2 = 230.1 KN-m

FOS = $M_{OR} / M_{OT}$ 3.36 > 1.5 ok

Check for soil bearing pressure (Operating Fluid Load): For Seismic Load

Seismic force at center of mass equipment = 12 KN

acting at a height of 3.800 m from BOF

$M_{OT}$ = Overturning moment = 45.6 KN-m

$M_{OT} / S_{xx}$ = 23.6 $KN/m^2$

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 72.7 $KN/m^2$

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 89.2 $KN/m^2$ < 375.0 $Kn/m^2$ OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 56.2 $KN/m^2$


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Check for differential settlement (Max Fluid Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = W_w/t_R + 0.7 \times P_{max} + h_R\gamma_c$ = 81.6 Kn/m$^2$ < 375.0 KN/m$^2$ Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 80.8 Kn/m$^2$ < 375.0 KN/m$^2$ Ok

As the tank encroaches the ring wall by 300mm out of total 450mm , the pressure under the tank is reduced to 300/450 = 0.7 times.

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 37.3 Kn/m

Hoop tension in ringwall, T = 1/2FD = 46.6 Kn

Factored design tension, $T_u$= 65.3 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 173 mm$^2$ $\phi$ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1620 mm$^2$

$A_{s,reqd}$ (per face of ring wall)= 810 mm$^2$

Required : 12 dia bar @ 252 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{s,\ provided}$(per face) = 1358 mm$^2$ ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 540 mm$^2$/m

$A_{s,reqd}$ (per face of ring wall)= 270 mm$^2$

Required : 10 dia bar @ 291 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

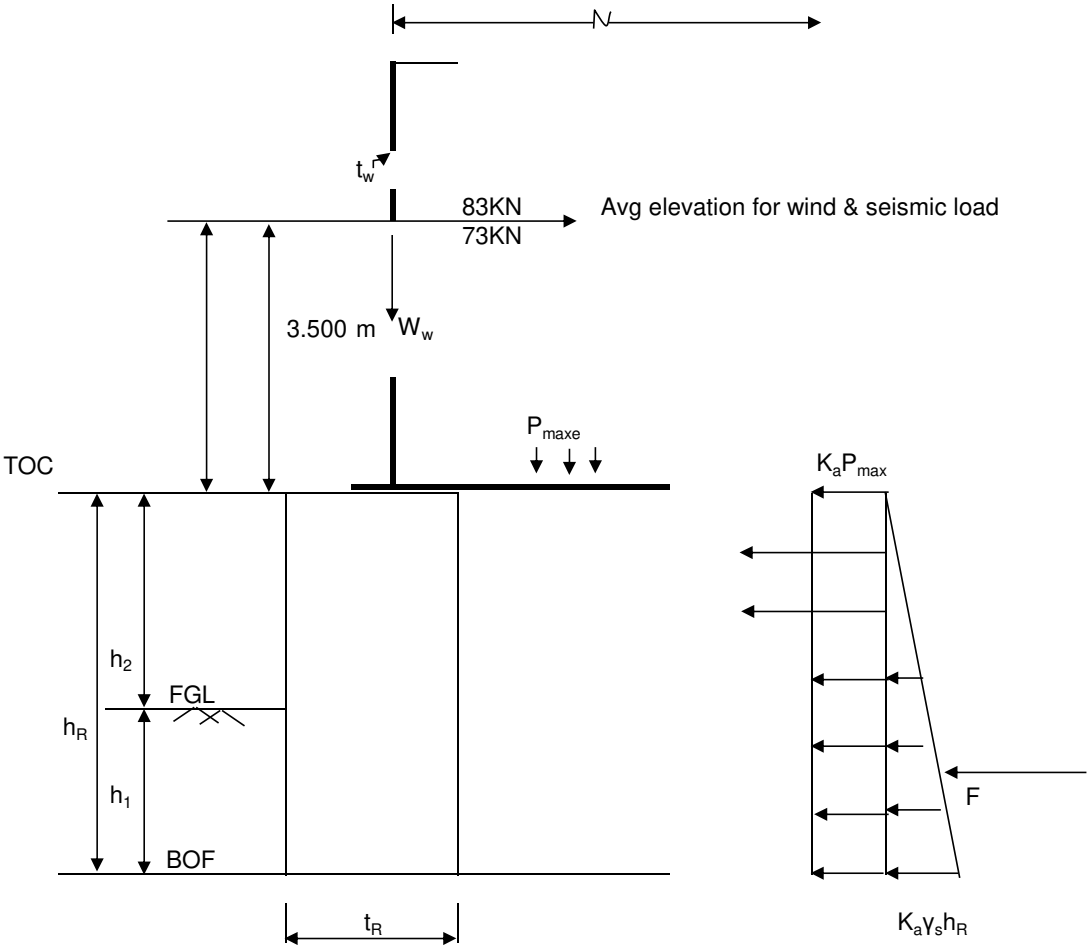
$A_{s,\ provided}$(per face) = 393 mm$^2$ ok


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Design Calculation of Foundation for 3650-TNK-006




Typical Section Through Tank

Pressure on ring wall


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Tank Data :

| | | |
|---|---|---|
| Diameter of tank ,D = | 5.5 | m |
| Self weight of tank = | 150 | KN |
| Self weight of tank per m of circumference,$W_w$ = | 8.68 | KN/m |
| Maximum weight of Fluid = | 1500 | KN |
| Pressure at base of tank ,$P_{max}$ | 63.1 | $KN/m^2$ |
| Design height of liquid, $h_L$ = | 6.7 | m |

Foundation Material :

| | | |
|---|---|---|
| Concrete, $f_c'$ = | 35 | Mpa |
| Reinforcement, $f_y$ = | 420 | Mpa |
| Unit weight of concrete,$\gamma_c$ = | 24 | $Kn/m^3$ |

Soil Data :

| | | |
|---|---|---|
| Unit weight of soil,$\gamma_s$ = | 20 | $Kn/m^3$ |
| Allowable bearing pressure, $q_{all}$= | 375 | $Kn/m^2$ |
| Coefficient of active earth pressure,$K_a$ = | 0.33 | |

Design Data of Ringwall :

| | | |
|---|---|---|
| Ringwall depth, $h_R = h_1 + h_2$ = | 1.8 | m |
| FGL to Bottom of foundation ,$h_1$ = | 1.2 | m |
| FGL to Top of concrete ,$h_2$ = | 0.6 | m |
| Ringwall thickness, $t_R$ = | 600 | mm |
| Inner dia of ringwall,$D_i$ = | 4.9 | m |
| Outer dia of ringwall,$D_o$ = | 6.1 | m |
| $S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ = | 13.01 | $m^3$ |


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Check of soil bearing pressure (Max Fluid Load): For wind Load

Wind force = 73 kn acting at a height of 5.300 m from BOF

$M_{OT}$ = Overturning moment = 386.9 KN-m

$M_{OT} / S_{xx}$ = 29.7 $KN/m^2$

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 89.2 $KN/m^2$

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 119 $KN/m^2$ < 375.0 $Kn/m^2$ OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 59.5 $KN/m^2$

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = W_w/t_R + h_R\gamma_c$ = 57.7 $KN/m^2$

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 87.4 $KN/m^2$

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 27.92 $KN/m^2$ no -ve pressure

Check for overturning (Empty Tank): For wind Load

$M_{OT}$ = Overturning moment = 386.9 KN-m

$M_{OR}$ = Restoring moment = (150+PI()*5.5*0.6*1.8*25)*5.5/2 = 1695.4 KN-m

FOS = $M_{OR} / M_{OT}$ 4.38 > 1.5 ok

Check for soil bearing pressure (Operating Fluid Load): For Seismic Load

Seismic force at center of mass equipment = 83 KN

acting at a height of 5.300 m from BOF

$M_{OT}$ = Overturning moment = 440 KN-m

$M_{OT} / S_{xx}$ = 33.8 $KN/m^2$

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 89.2 $KN/m^2$

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 113 $KN/m^2$ < 375.0 $Kn/m^2$ OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 65.6 $KN/m^2$


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Check for differential settlement (Max Fluid Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 97.1 Kn/m$^2$ < 375.0 KN/m$^2$ Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 99.1 Kn/m$^2$ < 375.0 KN/m$^2$ Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 48.2 Kn/m

Hoop tension in ringwall, T = 1/2FD = 139 Kn

Factored design tension, $T_u$= 194 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 514 mm$^2$ $\phi$ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 2160 mm$^2$

$A_{s,reqd}$ (per face of ring wall)= 1080 mm$^2$

Required : 12 dia bar @ 189 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{s,\ provided}$(per face) = 1358 mm$^2$ ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 720 mm$^2$/m

$A_{s,reqd}$ (per face of ring wall)= 360 mm$^2$

Required : 10 dia bar @ 218 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

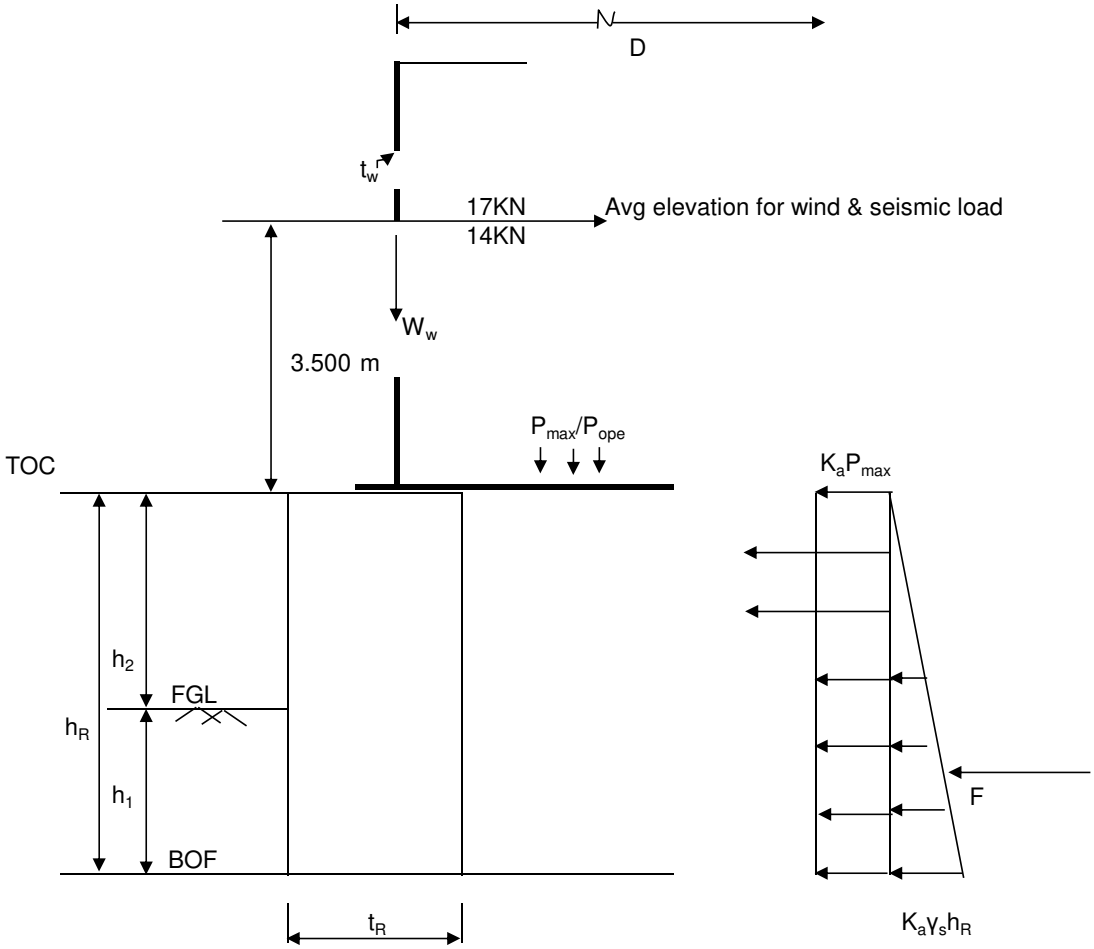
$A_{s,\ provided}$(per face) = 393 mm$^2$ ok


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Design Calculation of Foundation for 3820-TNK-001




Typical Section Through Tank

Pressure on ring wall


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Tank Data :

| | | |
|---|---|---|
| Diameter of tank ,D = | 3.34 | m |
| Self weight of tank = | 50.0 | KN |
| Self weight of tank per m of circumference,$W_w$ = | 4.77 | KN/m |
| Design height of liquid, $h_L$ = | 6.7 | m |
| Maximum weight of Fluid= | 330 | KN |
| Pressure at base of tank ,$P_{max}$ | 37.7 | $KN/m^2$ |
| Operating weight of Fluid= | 310 | KN |
| Pressure at base of tank ,$P_{ope}$ | 35.4 | $KN/m^2$ |

Foundation Material :

| | | |
|---|---|---|
| Concrete, $f_c'$ = | 35 | Mpa |
| Reinforcement, $f_y$ = | 420 | Mpa |
| Unit weight of concrete,$\gamma_c$ = | 24 | $Kn/m^3$ |

Soil Data :

| | | |
|---|---|---|
| Unit weight of soil,$\gamma_s$ = | 20 | $Kn/m^3$ |
| Allowable bearing pressure, $q_{all}$= | 375 | $Kn/m^2$ |
| Coefficient of active earth pressure,$K_a$ = | 0.33 | |

Design Data of Ringwall :

| | | |
|---|---|---|
| Ringwall depth, $h_R = h_1 + h_2$ = | 1.8 | m |
| FGL to Bottom of foundation ,$h_1$ = | 1.2 | m |
| FGL to Top of concrete ,$h_2$ = | 0.6 | m |
| Ringwall thickness, $t_R$ = | 600 | mm |
| Inner dia of ringwall,$D_i$ = | 2.74 | m |
| Outer dia of ringwall,$D_o$ = | 3.94 | m |
| $S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ = | 4.60 | $m^3$ |


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Check of soil bearing pressure (Max Fluid Load): For wind Load

Wind force = 14 kn acting at a height of 5.300 m from BOF

$M_{OT}$ = Overturning moment = 74.2 KN-m

$M_{OT} / S_{xx}$ = 16.1 $KN/m^2$

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 70 $KN/m^2$

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 86.1 $KN/m^2$ < 375.0 $Kn/m^2$ OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 53.8 $KN/m^2$

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = W_w/t_R + h_R\gamma_c$ = 51.1 $KN/m^2$

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 67.3 $KN/m^2$

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 35.01 $KN/m^2$ no -ve pressure

Check for overturning (Empty Tank): For wind Load

$M_{OT}$ = Overturning moment = 74.2 KN-m

$M_{OR}$ = Restoring moment = (50+PI()*3.34*0.6*1.8*25)*3.34/2 = 556.6 KN-m

FOS = $M_{OR} / M_{OT}$ 7.50 > 1.5 ok

Check for soil bearing pressure (Operating Fluid Load): For Seismic Load

Seismic force at center of mass equipment = 17 KN

acting at a height of 5.30 m from BOF

$M_{OT}$ = Overturning moment = 90.1 KN-m

$M_{OT} / S_{xx}$ = 19.6 $KN/m^2$

Load Combination : Dead + Fluid

$q_{RF} = W_w /t_R + P_{ope}/2 + h_R\gamma_c$ = 68.8 $KN/m^2$

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 82.5 $KN/m^2$ < 375.0 $Kn/m^2$ OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 55.1 $KN/m^2$


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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 70.0 Kn/m$^2$ < 375.0 KN/m$^2$ Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 73.7 Kn/m$^2$ < 375.0 KN/m$^2$ Ok

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 33.1 Kn/m

For Hydrotest :

Radial outward force on ringwall, $F = K_a h_R(\gamma_w h_L + \gamma_s h_R)$ = 50.5 Kn/m

Hoop tension in ringwall, T = 1/2FD = 84.3 Kn

Factored design tension, $T_u$ = 118 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 312 mm$^2$ $\phi$ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 2160 mm$^2$

$A_{s,reqd}$ (per face of ring wall)= 1080 mm$^2$

Required : 12 dia bar @ 189 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{s,\ provided}$(per face) = 1358 mm$^2$ ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 720 mm$^2$/m

$A_{s,reqd}$ (per face of ring wall)= 360 mm$^2$

Required : 10 dia bar @ 218 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 393 mm$^2$ ok


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## REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.


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## ATTACHMENT 1 INPUT DRAWINGS


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DRAWING NUMBER: MD-216-3650-EG-ME-GAD-1015 | REV: A01

PREVAILING WIND DIRECTION | PLANT NORTH | 32° | GEODETIC NORTH | 125° | MECCA | W | S | E

3650-025 HOLD — 3650-PMP-008 CARBON TRANSFER WATER PUMP
3650-PMP-011 CARBON FINES FILTER FEED PUMP — 3650-028 HOLD
3650-PMP-103 POX CARBON AREA SUMP PUMP — 3650-072 HOLD
E=264212.413 N=2506103.562
TOS EL. +913.550
B.O.B.(+)913.590 (TYP)
B.O.B.(+)913.300
B.O.B.(+)913.100
TOC EL. +912.540
3650-THK-001 COMBINED CARBON SETTLER — 3650-038 HOLD
3650-PMP-012 CARBON FINES FILTER FEED PUMP — 3650-029 HOLD
3650-REA-001 CARBON ATTRITIONING TANK — 3650-032 HOLD
ID OF TANK ⌀ 5500
ID OF SETTLER ⌀ 6000
ID OF REACTOR ⌀ 1800
ID OF COLUMN ⌀ 1200
3650-TNK-006 CARBON PROCESS WATER TANK — 3650-044 HOLD
3650-PMP-007 CARBON TRANSFER WATER PUMP — 3650-024 HOLD
TOC EL. +912.700
3650-PMP-001 POX ELUTION PUMP — 3650-018 HOLD
3650-PMP-009 POX REGENERATED CARBON PUMP — 3650-026 HOLD
3610-TNK-001 POX CIL WASH SOLUTION TANK
3610-PMP-013 POX CIL WASH SOLUTION PUMP — 3610-016 HOLD
3610-PMP-012 POX CIL WASH SOLUTION PUMP — 3610-015 HOLD
3650-TNK-002 POX ELUTION TANK — 3650-040 HOLD
3650-PMP-010 POX REGENERATED CARBON PUMP — 3650-027 HOLD
3650-PMP-102 POX CARBON AREA SUMP PUMP — 3650-031 HOLD
3820-PMP-003 HYDROCHLORIC ACID DOSING PUMP — 3820-007 HOLD
3820-PMP-005 HYDROCHLORIC ACID DOSING PUMP — 3820-009 HOLD
3650-PMP-002 POX ELUTION PUMP — 3650-019 HOLD
TOS EL. +913.350
3650-TNK-001 POX ACID WASH TANK — 3650-039 HOLD
3650-PMP-006 POX ACID WASH PUMP — 3650-023 HOLD
TOC EL. +912.470
3650-COL-001 POX ACID WASH COLUMN — 3650-002 HOLD
3650-COL-002 POX ELUTION COLUMN — 3650-003 HOLD
3820-TNK-001 HYDROCHLORIC ACID STORAGE TANK — 3820-012 HOLD
3820-PMP-004 HYDROCHLORIC ACID DOSING PUMP — 3820-008 HOLD
TOS EL. +913.850
3650-PMP-101 POX ACID WASH AREA SUMP PUMP — 3650-030 HOLD
3650-PMP-005 POX ACID WASH PUMP — 3650-022 HOLD
3650-TNK-003 POX NEUTRALIZATION WASH TANK — 3650-041 HOLD
E=264197.660 N=2506067.163

4300 | 4000 | 3150 | 1050 | 660 | 4200 | 4200 | 5200 | 4200 | 5700 | 31800

3650/1 | 3650/2 | 3650/3 | 3650/4 | 3650/5 | 3650/6 | 3650/7 | 3650/8

3650/F | 3650/E | 3650/D | 3650/C | 3650/B | 3650/A

4550 | 6800 | 2850 | 3900 | 4000 | 4000

PLAN AT EL.(+)912.700

NOTE:
FOR LOAD TABLE REFER DRAWING NUMBER MD-216-3650-EG-ME-GAD-1030

0 1 10m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

## GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED. 05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS: +Z, +Mz, +X, +Mx, +Y, +My

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM- GRATING 5 kN/m² — GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN.

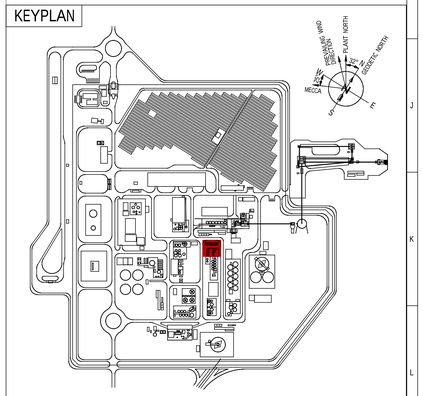
13 - REPORTED SEISMIC LOAD ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301-2007. HOWEVER, IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNER.

14 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.700 m.

15 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

16 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

## KEYPLAN



| DOCUMENT NUMBER | REFERENCE DOCUMENTS |
|---|---|
| MD-216-3000-EG-GM-LST-1010 | HOLD LIST |
| MD-216-3650-EG-ME-GAD-1030 | LOAD TABLE |
| MD-216-3650-EG-ME-GAD-1018 | LOAD PLAN, PLAN AT EL.+925.200 & EL.+927.400 |
| MD-216-3650-EG-ME-GAD-1017 | LOAD PLAN, PLAN AT EL.+918.700 & EL.+921.300 |
| MD-216-3650-EG-ME-GAD-1016 | LOAD PLAN, PLAN AT EL.+915.400 |
| MD-216-3650-EG-ME-GAD-1002 | GENERAL ARRANGEMENT DRAWING PLAN AT EL. +912.700 |

| REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|
| A01 | 14-04-2020 | ISSUED FOR APPROVAL | CK | TV | SC | RL | |

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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| | | | |
|---|---|---|---|
| LARSEN & TOUBRO DRAWING NUMBER | O19045_450_M_D_02975_001 | | REVISION 0 |
| OUTOTEC DRAWING NUMBER | 900707 _GAU01 _EE12 _MLD14 _36501 | DOCUMENT ID OU100680658 | REVISION 0 |
| DRAWN | CK | 14-04-2020 | MANSOURAH-MASSARAH GOLD PROJECT |
| CHECKED | TV | 14-04-2020 | LOAD PLAN |
| ENGINEER | SC | 14-04-2020 | POX ACID WASH/ELUTION/CARBON REGENERATION |
| APPROVED | RL | 14-04-2020 | PLAN AT EL. +912.700 |
| EMPLOYER | | | |
| SCALE 1:50 | A0 | DRAWING NUMBER MD-216-3650-EG-ME-GAD-1015 | REV A01 |

DRAWING NUMBER: MD-216-3650-EG-ME-GAD-1030 | REV: A01

| LOADING POINT | P1.1/.../P1.4 | | | | | | P2 | | | | | | P3 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 2.5 | | | | | | - 12 | | | | | | - 12 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | | | | | | | - 18 | | | | | | - 18 | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | - 17 | | | | | | | | | | | | | | | | | |
| 10 WIND ±X - DIRECTION | | 2.5 | | 2 | | | | | 2 | | | | | 2 | | | | |
| 11 WIND ±Y - DIRECTION | | 2.5 | | | 2 | | | | | 2 | | | | | 2 | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | 1 | | 1 | | | | | 2 | | | | | 2 | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | 1 | | | 1 | | | | | 2 | | | | | 2 | | | |
| EQUIPMENT - NO. | 3650-REA-001 | | | | | | 3650-PMP-009 | | | | | | 3650-PMP-010 | | | | | |
| DESCRIPTION | CARBON ATTRITIONING TANK | | | | | | POX REGENERATED CARBON PUMP | | | | | | POX REGENERATED CARBON PUMP | | | | | |

| LOADING POINT | P4 | | | | | | P5.1/.../5.4 | | | | | | P6.1/.../6.4 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 12 | | | | | | - 10 | | | | | | - 10 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | - 18 | | | | | | | | | | | | | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | | | | | | | - 50 | | | | | | - 50 | | | | | |
| 10 WIND ±X - DIRECTION | | | 2 | | | | 100 | 8 | | | | | 100 | 8 | | | | |
| 11 WIND ±Y - DIRECTION | | | | 2 | | | 100 | | 8 | | | | 100 | | 8 | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | 2 | | | | 57 | 4 | | | | | 57 | 4 | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | 2 | | | 57 | | | 4 | | | 57 | | | 4 | | |
| EQUIPMENT - NO. | 3650-PMP-103 | | | | | | 3650-COL-002 | | | | | | 3650-COL-001 | | | | | |
| DESCRIPTION | POX CARBON AREA SUMP PUMP | | | | | | POX ELUTION COLUMN | | | | | | POX ACID WASH COLUMN | | | | | |

| LOADING POINT | P7 | | | | | | P8 | | | | | | P9 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 25 | | | | | | - 12 | | | | | | - 25 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | | | | | | | - 18 | | | | | | | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | - 220 | | | | | | | | | | | | - 220 | | | | | |
| 10 WIND ±X - DIRECTION | | | 18 | | | | | 2 | | | | | | 18 | | | | |
| 11 WIND ±Y - DIRECTION | | | | 18 | | | | | 2 | | | | | | 18 | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | 12 | | | | | 2 | | | | | | 12 | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | 12 | | | | | 2 | | | | | | 12 | | | |
| EQUIPMENT - NO. | 3650-TNK-003 | | | | | | 3650-PMP-005 | | | | | | 3650-TNK-001 | | | | | |
| DESCRIPTION | POX NEUTRALIZATION WASH TANK | | | | | | POX ACID WASH PUMP | | | | | | POX ACID WASH TANK | | | | | |

| LOADING POINT | P10 | | | | | | P11 | | | | | | P12 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 12 | | | | | | - 12 | | | | | | - 50 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | - 18 | | | | | | - 18 | | | | | | | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | | | | | | | | | | | | | - 330 | | | | | |
| 10 WIND ±X - DIRECTION | | | 2 | | | | | 2 | | | | | | - | 14 | - | | |
| 11 WIND ±Y - DIRECTION | | | | 2 | | | | | 2 | | | | | - | - | 14 | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | 2 | | | | | 2 | | | | | | - | 17 | - | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | 2 | | | | | 2 | | | | | - | - | 17 | | |
| EQUIPMENT - NO. | 3650-PMP-006 | | | | | | 3650-PMP-101 | | | | | | 3820-TNK-001 | | | | | |
| DESCRIPTION | POX ACID WASH PUMP | | | | | | POX ACID WASH AREA SUMP PUMP | | | | | | HYDROCHLORIC ACID STORAGE TANK | | | | | |

| LOADING POINT | P13 | | | | | | P14 | | | | | | P15 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 3 | | | | | | - 3 | | | | | | - 3 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | - 5 | | | | | | - 5 | | | | | | - 5 | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | | | | | | | | | | | | | | | | | | |
| 10 WIND ±X - DIRECTION | - | 0.5 | - | | | | - | 0.5 | - | | | | - | 0.5 | - | | | |
| 11 WIND ±Y - DIRECTION | - | - | 0.5 | | | | - | - | 0.5 | | | | - | - | 0.5 | | | |
| 12 EARTHQUAKE ±X - DIRECTION | - | 0.2 | - | | | | - | 0.2 | - | | | | - | 0.2 | - | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | - | - | 0.2 | | | | - | - | 0.2 | | | | - | - | 0.2 | | | |
| EQUIPMENT - NO. | 3820-PMP-004 | | | | | | 3820-PMP-005 | | | | | | 3820-PMP-003 | | | | | |
| DESCRIPTION | HYDROCHLORIC ACID DOSING PUMP | | | | | | HYDROCHLORIC ACID DOSING PUMP | | | | | | HYDROCHLORIC ACID UNLOADING PUMP | | | | | |

| LOADING POINT | P16 | | | | | | P17.1/.../P17.3 | | | | | | P18 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 12 | | | | | | - 21 | | | | | | - 2 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | - 130 | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | | | | | | | | | | | | | - 4 | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | | | | | | | - 170 | | | | | | | | | | | |
| 10 WIND ±X - DIRECTION | | | 2 | | | | | 5 | | | | | | 0.5 | | | | |
| 11 WIND ±Y - DIRECTION | | | | 2 | | | | | 5 | | | | | | 0.5 | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | 2 | | | | | 3.5 | | | | | | 0.1 | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | 2 | | | | | 3.5 | | | | | | 0.1 | | | |
| EQUIPMENT - NO. | 3650-PMP-007 | | | | | | 3610-TNK-001 | | | | | | 3610-PMP-012 | | | | | |
| DESCRIPTION | CARBON TRANSFER WATER PUMP | | | | | | POX CIL TANK | | | | | | POX CIL WASH SOLUTION PUMP | | | | | |

| LOADING POINT | P19 | | | | | | P20 | | | | | | P21 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 2 | | | | | | - 150 | | | | | | - 12 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | - 4 | | | | | | | | | | | | | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | | | | | | | - 1500 | | | | | | - 18 | | | | | |
| 10 WIND ±X - DIRECTION | | 0.5 | | | | | | 73 | | | | | | 2 | | | | |
| 11 WIND ±Y - DIRECTION | | | 0.5 | | | | | | 73 | | | | | | 2 | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | 0.1 | | | | | | 83 | | | | | | 2 | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | 0.1 | | | | | | 83 | | | | | | 2 | | | |
| EQUIPMENT - NO. | 3610-PMP-013 | | | | | | 3650-TNK-006 | | | | | | 3650-PMP-007 | | | | | |
| DESCRIPTION | POX CIL WASH SOLUTION PUMP | | | | | | CARBON PROCESS WATER TANK | | | | | | CARBON TRANSFER WATER PUMP | | | | | |

| LOADING POINT | P22 | | | | | | P23.1/.../P23.8 | | | | | | P24 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 12 | | | | | | - 14 | | | | | | - 12 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | | | | | | | | | | | | | - 18 | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | - 18 | | | | | | - 170 | | | | | | | | | | | |
| 10 WIND ±X - DIRECTION | | | 2 | | | | | 22 | 9.5 | | | | | 2 | | | | |
| 11 WIND ±Y - DIRECTION | | | | 2 | | | 2 | 22 | | 9.5 | | | | | 2 | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | 2 | | | | | 25 | 11 | | | | | 2 | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | 2 | | | 2 | | | 11 | | | | | 2 | | | |
| EQUIPMENT - NO. | 3650-PMP-008 | | | | | | 3650-THK-001 | | | | | | 3650-PMP-012 | | | | | |
| DESCRIPTION | CARBON TRANSFER WATER PUMP | | | | | | COMBINED CARBON SETTLER | | | | | | CARBON FINES FILTER FEED PUMP | | | | | |

| LOADING POINT | P25 | | | | | | P26 | | | | | | P27 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 12 | | | | | | - 35 | | | | | | - 12 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | - 18 | | | | | | | | | | | | - 18 | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | | | | | | | - 450 | | | | | | | | | | | |
| 10 WIND ±X - DIRECTION | | | 2 | | | | | 22 | | | | | | 2 | | | | |
| 11 WIND ±Y - DIRECTION | | | | 2 | | | | | 22 | | | | | | 2 | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | 2 | | | | | 24 | | | | | | 2 | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | 2 | | | | | 24 | | | | | | 2 | | | |
| EQUIPMENT - NO. | 3650-PMP-011 | | | | | | 3650-TNK-002 | | | | | | 3650-PMP-002 | | | | | |
| DESCRIPTION | CARBON FINES FILTER FEED PUMP | | | | | | POX ELUTION TANK | | | | | | POX ELUTION PUMP | | | | | |

| LOADING POINT | P28 | | | | | | P29.1/.../29.8 | | | | | | P30 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 12 | | | | | | - 30 | | | | | | - 10 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | - 18 | | | | | | | | | | | | | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | | | | | | | - 35 | | | | | | - 15 | | | | | |
| 10 WIND ±X - DIRECTION | | | 2 | | | | | | | | | | | | | | | |
| 11 WIND ±Y - DIRECTION | | | | 2 | | | | | | | | | | | | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | 2 | | | | | | | | | | | | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | 2 | | | | | | | | | | | | | | |
| EQUIPMENT - NO. | 3650-PMP-001 | | | | | | 3650-FLT-001 | | | | | | 3650-HEX-005 | | | | | |
| DESCRIPTION | POX ELUTION PUMP | | | | | | CARBON FINES FILTER | | | | | | POX HEATER HEAT EXCHANGER | | | | | |

| LOADING POINT | P31 | | | | | | P32 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 10 | | | | | | - 10 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | | | | | | | | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | |
| 9 LIVE LOAD | - 15 | | | | | | - 15 | | | | | |
| 10 WIND ±X - DIRECTION | | | | | | | | | | | | |
| 11 WIND ±Y - DIRECTION | | | | | | | | | | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | | | | | | | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | | | | | | | | | |
| EQUIPMENT - NO. | 3650-HEX-004 | | | | | | 3650-HEX-003 | | | | | |
| DESCRIPTION | POX HEATER HEAT EXCHANGER | | | | | | POX EVALUATE HEAT EXCHANGER | | | | | |

| LOADING POINT | P33 | | | | | | P34 | | | | | | P35 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 10 | | | | | | - 10 | | | | | | - 5 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | | | | | | | | | | | | | - 8 | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | - 15 | | | | | | - 15 | | | | | | | | | | | |
| 10 WIND ±X - DIRECTION | | | | | | | | | | 1 | | | | | 2 | | | |
| 11 WIND ±Y - DIRECTION | | | | | | | | | | | 1 | | | | | 2 | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | | | | | | | | 0.3 | | | | | 1.5 | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | | | | | | | | 0.3 | | | | | 1.5 | | |
| EQUIPMENT - NO. | 3650-HEX-002 | | | | | | 3650-HEX-001 | | | | | | 3820-FAN-001 | | | | | |
| DESCRIPTION | POX RECLAIM HEAT EXCHANGER | | | | | | POX RECLAIM HEAT EXCHANGER | | | | | | HYDROCHLORIC ACID AREA OFFGAS FAN | | | | | |

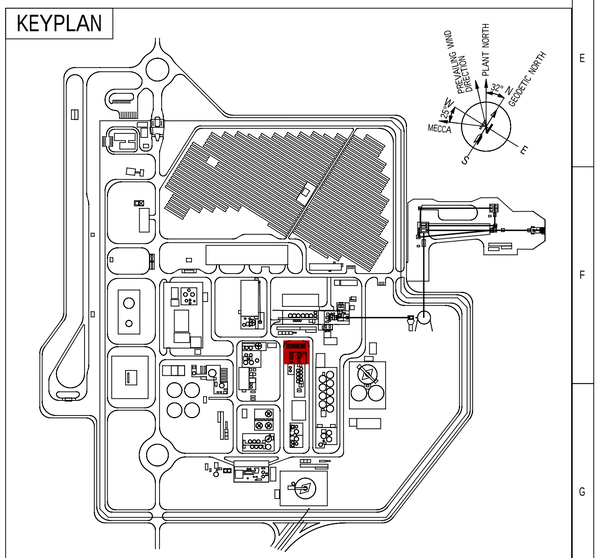
| LOADING POINT | P36.1/.../P36.12 | | | | | | P37.1/.../37.4 | | | | | | P38.1/.../38.4 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 2 | | | | | | - 4 | | | | | | - 35 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | - 3 | | | | | | | | | | | | | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | | | | | | | - 40 | | | | | | - 45 | | | | | |
| 10 WIND ±X - DIRECTION | | | 2 | | | | | 14 | | | | | | 12 | | | | |
| 11 WIND ±Y - DIRECTION | | | | 1 | | | | | 14 | | | | | | 12 | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | 1.5 | | | | | 2 | | | | | | 12 | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | 1.5 | | | | | 2 | | | | | | 12 | | | |
| EQUIPMENT - NO. | 3650-SCR-001 | | | | | | 3650-TNK-005 | | | | | | 3650-FNC-001 | | | | | |
| DESCRIPTION | POX CARBON DEWATERING SCREEN | | | | | | POX QUENCH TANK | | | | | | POX REGENERATION KILN | | | | | |

| LOADING POINT | P40.1/.../40.4 | | | | | | P41.1/.../41.4 | | | | | | P42 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 25 | | | | | | - 5 | | | | | | - 5 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | | | | | | | | | | | | | - 8 | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | - 45 | | | | | | - 10 | | | | | | | | | | | |
| 10 WIND ±X - DIRECTION | | | 3 | | | | | 3 | | | | | | 1 | | | | |
| 11 WIND ±Y - DIRECTION | | | | 25 | | | | | 3 | | | | | | 1 | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | 2.5 | | | | | 2.5 | | | | | | 0.3 | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | 12 | | | | | 2.5 | | | | | | 0.3 | | | |
| EQUIPMENT - NO. | 3650-HOP-001 | | | | | | 3650-SCR-002 | | | | | | 3650-SCB-001 | | | | | |
| DESCRIPTION | POX REGENERATION KILN FEED HOPPER | | | | | | TRASH STATIC SCREEN | | | | | | HYDROCHLORIC ACID AREA OFFGASDEMISTER | | | | | |

| LOADING POINT | P43.1/.../P43.2 | | | | | | P44.1/.../P44.2 | | | | | | P45.1/.../P45.2 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 9 | | | | | | - 5 | | | | | | - 7 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | | | | | | | | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | | | | | | | | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | | | | | | | | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | | | | | | | | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | | | | | | | | | | | | | | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | | | | | | | | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | | | | | | | | | | | | | |
| 9 LIVE LOAD | | | | | | | - 10 | | | | | | - 18 | | | | | |
| 10 WIND ±X - DIRECTION | | | 13 | | | | | | | | | | | | | | | |
| 11 WIND ±Y - DIRECTION | | | | 13 | | | | | | | | | | | | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | 9 | | | | | | | | | | | | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | 9 | | | | | | | | | | | | | | |
| EQUIPMENT - NO. | 3650-CHT-004 | | | | | | 3650-FDB-001 | | | | | | 3650-CHT-002 | | | | | |
| DESCRIPTION | POX CARBON DE- WATERING SCREEN OVERSIZE CHUTE | | | | | | POX CARBON DE- WATERING SCREEN FEED BOX | | | | | | POX CARBON DE- WATERING SCREEN UNDERPAN | | | | | |

| LOADING POINT | P46.1/.../P46.2 | | | | | |
|---|---|---|---|---|---|---|
| LOAD CASE | Vz[kN] | Hx[kN] | | Hy[kN] | | |
| 1 DEAD LOAD EQUIPMENT | - 5 | | | | | |
| 2 DEAD LOAD (SECONDARY EQUIPMENT) | | | | | | |
| 3 OPERATION LOAD 1. FILLING | | | | | | |
| 4 OPERATION LOAD 2. PRESSURE | | | | | | |
| 5 OPERATION LOAD 3. TEMPERATURE | | | | | | |
| 6 OPERATION LOAD 4. DYNAMIC | | | | | | |
| 7 OPERATION LOAD 5. OTHERS | | | | | | |
| 8 TEST LOAD FILLING / EMERGENCY | | | | | | |
| 9 LIVE LOAD | - 10 | | | | | |
| 10 WIND ±X - DIRECTION | | | | | | |
| 11 WIND ±Y - DIRECTION | | | | | | |
| 12 EARTHQUAKE ±X - DIRECTION | | | | | | |
| 13 EARTHQUAKE ±Y - DIRECTION | | | | | | |
| EQUIPMENT - NO. | 3650-CHT-001 | | | | | |
| DESCRIPTION | POX CARBON DE- WATERING SCREEN OVERSIZE CHUTE | | | | | |

KEYPLAN




| DOCUMENT NUMBER | REFERENCE DOCUMENTS | REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|---|---|
| MD-216-3000-EG-GM-LST-1010 | HOLD LIST | | | | | | | | |
| MD-216-3650-EG-ME-GAD-1018 | LOAD PLAN, PLAN AT EL.+925.200 & EL.+927.400 | | | | | | | | |
| MD-216-3650-EG-ME-GAD-1017 | LOAD PLAN, PLAN AT EL.+918.700 & EL.+921.300 | | | | | | | | |
| MD-216-3650-EG-ME-GAD-1016 | LOAD PLAN, PLAN AT EL.+915.400 | | | | | | | | |
| MD-216-3650-EG-ME-GAD-1015 | LOAD PLAN, PLAN AT EL.+912.700 | | | | | | | | |
| MD-216-3650-EG-ME-GAD-1002 | GENERAL ARRANGEMENT DRAWING PLAN AT EL. +912.700 | A01 | 14-04-2020 | ISSUED FOR APPROVAL | NA | TV | SC | RL | |

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Outotec

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_M_D_02918_059 | REVISION: 0

OUTOTEC DRAWING NUMBER: 900707 _GAU01 _EE12 _MLD14 _36505 | DOCUMENT ID: OU100681424 | REVISION: 0

| | | |
|---|---|---|
| DRAWN | NA | 14-04-2020 |
| CHECKED | TV | 14-04-2020 |
| ENGINEER | SC | 14-04-2020 |
| APPROVED | RL | 14-04-2020 |
| EMPLOYER | | |
| SCALE | - | A1 |

MANSOURAH-MASSARAH GOLD PROJECT
LOAD PLAN
POX ACID WASH/ELUTION/CARBON REGENERATION LOAD TABLE

DRAWING NUMBER: MD-216-3650-EG-ME-GAD-1030 | REV: A01

# ATTACHMENT 2 STAAD INPUT - FDN OF 3650-COL-001 & 002


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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18.05.2020
JOB NAME MMGP/3650-COL-001 & 002
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
CHECKER NAME SJM
APPROVED NAME AG
END JOB INFORMATION
UNIT METER KN
JOINT COORDINATES
7783 365.388 912.75 -383.293; 7787 365.388 911 -383.293;
7968 366.675 911 -383.293; 7969 366.675 912.75 -383.293;
7971 366.031 911 -382.65; 7972 366.031 912.75 -382.65;
7973 366.031 911 -383.937; 7974 366.031 912.75 -383.937;
7975 366.031 911 -382.293; 7976 366.031 911 -384.293;
7977 365.031 911 -383.293; 7978 367.031 911 -383.293;
7979 367.031 911 -382.293; 7980 367.031 911 -384.293;
7981 365.031 911 -382.293; 7982 365.031 911 -384.293;
7983 365.388 911 -384.293; 7984 365.388 911 -382.293;
7985 366.675 911 -384.293; 7986 366.675 911 -382.293;
7987 365.71 911 -384.293; 7988 366.353 911 -384.293;
7989 365.71 911 -382.293; 7990 366.353 911 -382.293;
7991 365.71 911 -383.293; 7992 366.031 911 -383.293;
7993 366.353 911 -383.293; 7994 365.71 911 -382.65;
7995 365.388 911 -382.65; 7996 365.031 911 -382.65;
7997 366.353 911 -382.65; 7998 366.675 911 -382.65;
7999 367.031 911 -382.65; 8000 365.71 911 -383.937;
8001 365.388 911 -383.937; 8002 365.031 911 -383.937;
8003 366.353 911 -383.937; 8004 366.675 911 -383.937;
8005 367.031 911 -383.937; 8006 365.031 911 -383.615;
8007 365.031 911 -382.971; 8008 367.031 911 -383.615;
8009 367.031 911 -382.971; 8010 365.388 911 -382.971;
8011 366.031 911 -383.615; 8012 366.675 911 -383.615;
8013 366.675 911 -382.971; 8014 365.71 911 -382.971;
8015 366.031 911 -382.971; 8016 366.353 911 -382.971;
8017 365.71 911 -383.615; 8018 365.388 911 -383.615;
8019 366.353 911 -383.615; 8020 367.281 911 -383.293;
8021 367.281 911 -382.293; 8022 367.281 911 -384.293;
8023 367.281 911 -382.65; 8024 367.281 911 -383.937;
8025 367.281 911 -383.615; 8026 367.281 911 -382.971;
8027 364.781 911 -383.293; 8028 364.781 911 -382.293;
8029 364.781 911 -384.293; 8030 364.781 911 -382.65;
8031 364.781 911 -383.937; 8032 364.781 911 -383.615;
8033 364.781 911 -382.971; 8034 366.031 911 -382.043;
8035 367.031 911 -382.043; 8036 365.031 911 -382.043;
8037 365.388 911 -382.043; 8038 366.675 911 -382.043;
8039 365.71 911 -382.043; 8040 366.353 911 -382.043;
8041 367.281 911 -382.043; 8042 364.781 911 -382.043;
8043 366.031 911 -384.543; 8044 367.031 911 -384.543;
8045 365.031 911 -384.543; 8046 365.388 911 -384.543;
8047 366.675 911 -384.543; 8048 365.71 911 -384.543;
8049 366.353 911 -384.543; 8050 367.281 911 -384.543;
8051 364.781 911 -384.543;
MEMBER INCIDENCES
4 7787 7783; 190 7968 7969; 193 7971 7972; 194 7973 7974;
ELEMENT INCIDENCES SHELL
279 7982 7983 8001 8002; 280 7983 7987 8000 8001;
281 7987 7976 7973 8000; 282 7976 7988 8003 7973;
283 7988 7985 8004 8003; 284 7985 7980 8005 8004;
285 8002 8001 8018 8006; 286 8001 8000 8017 8018;
287 8000 7973 8011 8017; 288 7973 8003 8019 8011;
289 8003 8004 8012 8019; 290 8004 8005 8008 8012;
291 8006 8018 7787 7977; 292 8018 8017 7991 7787;
293 8017 8011 7992 7991; 294 8011 8019 7993 7992;
295 8019 8012 7968 7993; 296 8012 8008 7978 7968;
297 7977 7787 8010 8007; 298 7787 7991 8014 8010;
299 7991 7992 8015 8014; 300 7992 7993 8016 8015;
301 7993 7968 8013 8016; 302 7968 7978 8009 8013;
303 8007 8010 7995 7996; 304 8010 8014 7994 7995;
305 8014 8015 7971 7994; 306 8015 8016 7997 7971;
307 8016 8013 7998 7997; 308 8013 8009 7999 7998;
309 7996 7995 7984 7981; 310 7995 7994 7989 7984;
311 7994 7971 7975 7989; 312 7971 7997 7990 7975;
313 7997 7998 7986 7990; 314 7998 7999 7979 7986;
315 7980 8022 8024 8005; 316 8005 8024 8025 8008;
317 8008 8025 8020 7978; 318 7978 8020 8026 8009;
```





```
317 8008 8025 8020 7978; 318 7978 8020 8026 8009;
319 8009 8026 8023 7999; 320 7999 8023 8021 7979;
321 8029 7982 8002 8031; 322 8031 8002 8006 8032;
323 8032 8006 7977 8027; 324 8027 7977 8007 8033;
325 8033 8007 7996 8030; 326 8030 7996 7981 8028;
327 8051 8045 7982 8029; 328 8045 8046 7983 7982;
329 8046 8048 7987 7983; 330 8048 8043 7976 7987;
331 8043 8049 7988 7976; 332 8049 8047 7985 7988;
333 8047 8044 7980 7985; 334 8044 8050 8022 7980;
335 8028 7981 8036 8042; 336 7981 7984 8037 8036;
337 7984 7989 8039 8037; 338 7989 7975 8034 8039;
339 7975 7990 8040 8034; 340 7990 7986 8038 8040;
341 7986 7979 8035 8038; 342 7979 8021 8041 8035;
START GROUP DEFINITION
ELEMENT
_FDN_SHEAR 279 280 283 TO 286 289 290 293 294 299 300 303 304 307 -
308 TO 310 313 TO 342
_FDN_MOMENT 279 TO 342
END GROUP DEFINITION
ELEMENT PROPERTY
279 TO 342 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
4 190 193 194 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
279 TO 342 PLATE MAT DIRECT Y SUBGRADE 48000 PRINT
*********************************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
7783 7969 7972 7974 FY -10
ELEMENT LOAD
* LOAD DUE TO BACKFILL
279 TO 342 PR 16
* LOAD DUE TO SURCHARGE
279 TO 342 PR 20
*********************************************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
7783 7969 7972 7974 FY -50
*********************************************************************************************************************
LOAD 3 LOADTYPE None  TITLE FLUID LOAD ( F )
JOINT LOAD
7783 7969 7972 7974 FY -0.0001
*********************************************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD ( W )
JOINT LOAD
7783 7969 7972 7974 FX 8 FY -100 FZ 8
*********************************************************************************************************************
LOAD 5 LOADTYPE None  TITLE SEISMIC LOAD ( E )
JOINT LOAD
7783 7969 7972 7974 FX 4 FY -57 FZ 4
*********************************************************************************************************************
* FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
*************************************************************************************************
LOAD COMB 101 1.4D + 1.4F
1 1.4 3 1.4
LOAD COMB 102 1.2D + 1.2F + 1.6L
1 1.2 3 1.2 2 1.6
LOAD COMB 103 1.2D + 1.0E + 0.5L
1 1.2 5 1.0 2 0.5
LOAD COMB 104 0.9D + 1.0E
1 0.9 5 1.0
LOAD COMB 105 0.9D + 1.6W
1 0.9 4 1.6
LOAD COMB 106 1.2D + 1.6W + 0.5L
1 1.2 4 1.6 2 0.5
*********************************************************************************************************************
```

```
*************************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
LOAD COMB 201 D + F
1 1.0 3 1.0
LOAD COMB 202 D + F + L
1 1.0 2 1.0 3 1.0
LOAD COMB 203 D + F + W
1 1.0 3 1.0 4 1.0
LOAD COMB 204 0.6D + W
1 0.6 4 1.0
LOAD COMB 205 0.6D + 0.7E
1 0.6 5 0.7
LOAD COMB 206 D + 0.75W + 0.75L
1 1.0 4 0.75 2 0.75
*************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 106
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 4 190 193 194
CLB 0.05 MEMB 4 190 193 194
CLS 0.05 ALL
CLT 0.05 MEMB 4 190 193 194
MAXMAIN 20 ALL
MINMAIN 20 ALL
MINSEC 10 ALL
DESIGN COLUMN 4 190 193 194
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18.05.2020
JOB NAME MMGP/3650-COL-001 & 002_STABILITY
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
CHECKER NAME SJM
APPROVED NAME AG
END JOB INFORMATION
UNIT METER KN
JOINT COORDINATES
7783 365.388 912.75 -383.293; 7787 365.388 911 -383.293;
7968 366.675 911 -383.293; 7969 366.675 912.75 -383.293;
7971 366.031 911 -382.65; 7972 366.031 912.75 -382.65;
7973 366.031 911 -383.937; 7974 366.031 912.75 -383.937;
MEMBER INCIDENCES
4 7787 7783; 190 7968 7969; 193 7971 7972; 194 7973 7974;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
4 190 193 194 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
7787 7968 7971 7973 FIXED
*********************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
7783 7969 7972 7974 FY -10
*********************************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
7783 7969 7972 7974 FY -50
*********************************************************************************************************
LOAD 3 LOADTYPE None  TITLE FLUID LOAD ( F )
JOINT LOAD
7783 7969 7972 7974 FY -0.0001
*********************************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD ( W )
JOINT LOAD
7783 7969 7972 7974 FX 8 FY -100 FZ 8
*********************************************************************************************************
LOAD 5 LOADTYPE None  TITLE SEISMIC LOAD ( E )
JOINT LOAD
7783 7969 7972 7974 FX 4 FY -57 FZ 4
*********************************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE & STABILITY
LOAD COMB 201 D + F
1 1.0 3 1.0
LOAD COMB 202 D + F + L
1 1.0 2 1.0 3 1.0
LOAD COMB 203 D + F + W
1 1.0 3 1.0 4 1.0
LOAD COMB 204 0.6D + W
1 0.6 4 1.0
LOAD COMB 205 0.6D + 0.7E
1 0.6 5 0.7
LOAD COMB 206 D + 0.75W + 0.75L
1 1.0 4 0.75 2 0.75
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

# ATTACHMENT 3 STAAD OUTPUT - FDN OF 3650-COL-001 & 002


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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    MAY 26, 2020                 *
*          Time=    16:49:50                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: 3650-COL-001 _002 - NEW.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 18.05.2020
    4. JOB NAME MMGP/3650-COL-001 _002
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. CHECKER NAME SJM
   10. APPROVED NAME AG
   11. END JOB INFORMATION
   12. UNIT METER KN
   13. JOINT COORDINATES
   14. 7783 365.388 912.75 -383.293; 7787 365.388 911 -383.293
   15. 7968 366.675 911 -383.293; 7969 366.675 912.75 -383.293
   16. 7971 366.031 911 -382.65; 7972 366.031 912.75 -382.65
   17. 7973 366.031 911 -383.937; 7974 366.031 912.75 -383.937
   18. 7975 366.031 911 -382.293; 7976 366.031 911 -384.293
   19. 7977 365.031 911 -383.293; 7978 367.031 911 -383.293
   20. 7979 367.031 911 -382.293; 7980 367.031 911 -384.293
   21. 7981 365.031 911 -382.293; 7982 365.031 911 -384.293
   22. 7983 365.388 911 -384.293; 7984 365.388 911 -382.293
   23. 7985 366.675 911 -384.293; 7986 366.675 911 -382.293
   24. 7987 365.71 911 -384.293; 7988 366.353 911 -384.293
   25. 7989 365.71 911 -382.293; 7990 366.353 911 -382.293
   26. 7991 365.71 911 -383.293; 7992 366.031 911 -383.293
   27. 7993 366.353 911 -383.293; 7994 365.71 911 -382.65
   28. 7995 365.388 911 -382.65; 7996 365.031 911 -382.65
   29. 7997 366.353 911 -382.65; 7998 366.675 911 -382.65
   30. 7999 367.031 911 -382.65; 8000 365.71 911 -383.937
   31. 8001 365.388 911 -383.937; 8002 365.031 911 -383.937
   32. 8003 366.353 911 -383.937; 8004 366.675 911 -383.937
   33. 8005 367.031 911 -383.937; 8006 365.031 911 -383.615
   34. 8007 365.031 911 -382.971; 8008 367.031 911 -383.615
   35. 8009 367.031 911 -382.971; 8010 365.388 911 -382.971
   36. 8011 366.031 911 -383.615; 8012 366.675 911 -383.615
   37. 8013 366.675 911 -382.971; 8014 365.71 911 -382.971
   38. 8015 366.031 911 -382.971; 8016 366.353 911 -382.971
```





```
   39. 8017 365.71 911 -383.615; 8018 365.388 911 -383.615
   40. 8019 366.353 911 -383.615; 8020 367.281 911 -383.293
   41. 8021 367.281 911 -382.293; 8022 367.281 911 -384.293
   42. 8023 367.281 911 -382.65; 8024 367.281 911 -383.937
   43. 8025 367.281 911 -383.615; 8026 367.281 911 -382.971
   44. 8027 364.781 911 -383.293; 8028 364.781 911 -382.293
   45. 8029 364.781 911 -384.293; 8030 364.781 911 -382.65
   46. 8031 364.781 911 -383.937; 8032 364.781 911 -383.615
   47. 8033 364.781 911 -382.971; 8034 366.031 911 -382.043
   48. 8035 367.031 911 -382.043; 8036 365.031 911 -382.043
   49. 8037 365.388 911 -382.043; 8038 366.675 911 -382.043
   50. 8039 365.71 911 -382.043; 8040 366.353 911 -382.043
   51. 8041 367.281 911 -382.043; 8042 364.781 911 -382.043
   52. 8043 366.031 911 -384.543; 8044 367.031 911 -384.543
   53. 8045 365.031 911 -384.543; 8046 365.388 911 -384.543
   54. 8047 366.675 911 -384.543; 8048 365.71 911 -384.543
   55. 8049 366.353 911 -384.543; 8050 367.281 911 -384.543
   56. 8051 364.781 911 -384.543
   57. MEMBER INCIDENCES
   58. 4 7787 7783; 190 7968 7969; 193 7971 7972; 194 7973 7974
   59. ELEMENT INCIDENCES SHELL
   60. 279 7982 7983 8001 8002; 280 7983 7987 8000 8001
   61. 281 7987 7976 7973 8000; 282 7976 7988 8003 7973
   62. 283 7988 7985 8004 8003; 284 7985 7980 8005 8004
   63. 285 8002 8001 8018 8006; 286 8001 8000 8017 8018
   64. 287 8000 7973 8011 8017; 288 7973 8003 8019 8011
   65. 289 8003 8004 8012 8019; 290 8004 8005 8008 8012
   66. 291 8006 8018 7787 7977; 292 8018 8017 7991 7787
   67. 293 8017 8011 7992 7991; 294 8011 8019 7993 7992
   68. 295 8019 8012 7968 7993; 296 8012 8008 7978 7968
   69. 297 7977 7787 8010 8007; 298 7787 7991 8014 8010
   70. 299 7991 7992 8015 8014; 300 7992 7993 8016 8015
   71. 301 7993 7968 8013 8016; 302 7968 7978 8009 8013
   72. 303 8007 8010 7995 7996; 304 8010 8014 7994 7995
   73. 305 8014 8015 7971 7994; 306 8015 8016 7997 7971
   74. 307 8016 8013 7998 7997; 308 8013 8009 7999 7998
   75. 309 7996 7995 7984 7981; 310 7995 7994 7989 7984
   76. 311 7994 7971 7975 7989; 312 7971 7997 7990 7975
   77. 313 7997 7998 7986 7990; 314 7998 7999 7979 7986
   78. 315 7980 8022 8024 8005; 316 8005 8024 8025 8008
   79. 317 8008 8025 8020 7978; 318 7978 8020 8026 8009
   80. 319 8009 8026 8023 7999; 320 7999 8023 8021 7979
   81. 321 8029 7982 8002 8031; 322 8031 8002 8006 8032
   82. 323 8032 8006 7977 8027; 324 8027 7977 8007 8033
   83. 325 8033 8007 7996 8030; 326 8030 7996 7981 8028
   84. 327 8051 8045 7982 8029; 328 8045 8046 7983 7982
   85. 329 8046 8048 7987 7983; 330 8048 8043 7976 7987
   86. 331 8043 8049 7988 7976; 332 8049 8047 7985 7988
   87. 333 8047 8044 7980 7985; 334 8044 8050 8022 7980
   88. 335 8028 7981 8036 8042; 336 7981 7984 8037 8036
   89. 337 7984 7989 8039 8037; 338 7989 7975 8034 8039
   90. 339 7975 7990 8040 8034; 340 7990 7986 8038 8040
   91. 341 7986 7979 8035 8038; 342 7979 8021 8041 8035
   92. START GROUP DEFINITION
   93. ELEMENT
   94. _FDN_SHEAR 279 280 283 TO 286 289 290 293 294 299 300 303 304 307 -
```

```
 95. 308 TO 310 313 TO 342
 96. _FDN_MOMENT 279 TO 342
 97. END GROUP DEFINITION
 98. ELEMENT PROPERTY
 99. 279 TO 342 THICKNESS 0.4
100. DEFINE MATERIAL START
101. ISOTROPIC CONCRETE
102. E 2.17185E+007
103. POISSON 0.17
104. DENSITY 25
105. ALPHA 1E-005
106. DAMP 0.05
107. TYPE CONCRETE
108. STRENGTH FCU 27579
109. END DEFINE MATERIAL
110. MEMBER PROPERTY AMERICAN
111. 4 190 193 194 PRIS YD 0.5 ZD 0.5
112. CONSTANTS
113. MATERIAL CONCRETE ALL
114. SUPPORTS
115. 279 TO 342 PLATE MAT DIRECT Y SUBGRADE 48000 PRINT

     JOINT      INFLUENCE AREA
     -----      --------------

      7787          0.109316
      7968          0.109157
      7971          0.108992
      7973          0.108988
      7975          0.097576
      7976          0.097417
      7977          0.097727
      7978          0.097567
      7979          0.091960
      7980          0.091810
      7981          0.092111
      7982          0.091960
      7983          0.102866
      7984          0.103034
      7985          0.102716
      7986          0.102884
      7987          0.097413
      7988          0.097567
      7989          0.097573
      7990          0.097727
      7991          0.103522
      7992          0.103526
      7993          0.103685
      7994          0.108988
      7995          0.115089
      7996          0.102887
      7997          0.109161
      7998          0.114921
      7999          0.102719
      8000          0.108984
      8001          0.115085
```





```
      8002          0.102884
      8003          0.109157
      8004          0.114916
      8005          0.102716
      8006          0.097727
      8007          0.097573
      8008          0.097567
      8009          0.097413
      8010          0.109144
      8011          0.103526
      8012          0.109157
      8013          0.108984
      8014          0.103358
      8015          0.103362
      8016          0.103522
      8017          0.103522
      8018          0.109316
      8019          0.103685
      8020          0.040251
      8021          0.037938
      8022          0.037876
      8023          0.042377
      8024          0.042375
      8025          0.040251
      8026          0.040188
      8027          0.040251
      8028          0.037938
      8029          0.037876
      8030          0.042377
      8031          0.042375
      8032          0.040251
      8033          0.040188
      8034          0.040185
      8035          0.037872
      8036          0.037934
      8037          0.042433
      8038          0.042371
      8039          0.040184
      8040          0.040247
      8041          0.015624
      8042          0.015624
      8043          0.040189
      8044          0.037876
      8045          0.037938
      8046          0.042437
      8047          0.042375
      8048          0.040188
      8049          0.040251
      8050          0.015626
      8051          0.015626
     TOTAL          6.249949
116. ********************************************************************************
117. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
118. SELFWEIGHT Y -1
119. JOINT LOAD
120. 7783 7969 7972 7974 FY -10
```

```
121. ELEMENT LOAD
122. * LOAD DUE TO BACKFILL
123. 279 TO 342 PR 16
124. * LOAD DUE TO SURCHARGE
125. 279 TO 342 PR 20
126. ********************************************************************************
127. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
128. JOINT LOAD
129. 7783 7969 7972 7974 FY -50
130. ********************************************************************************
131. LOAD 3 LOADTYPE NONE  TITLE FLUID LOAD ( F )
132. JOINT LOAD
133. 7783 7969 7972 7974 FY -0.0001
134. ********************************************************************************
135. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD ( W )
136. JOINT LOAD
137. 7783 7969 7972 7974 FX 8 FY -100 FZ 8
138. ********************************************************************************
139. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD ( E )
140. JOINT LOAD
141. 7783 7969 7972 7974 FX 4 FY -57 FZ 4
142. ********************************************************************************
143. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
144. ********************************************************************************
145. LOAD COMB 101 1.4D + 1.4F
146. 1 1.4 3 1.4
147. LOAD COMB 102 1.2D + 1.2F + 1.6L
148. 1 1.2 3 1.2 2 1.6
149. LOAD COMB 103 1.2D + 1.0E + 0.5L
150. 1 1.2 5 1.0 2 0.5
151. LOAD COMB 104 0.9D + 1.0E
152. 1 0.9 5 1.0
153. LOAD COMB 105 0.9D + 1.6W
154. 1 0.9 4 1.6
155. LOAD COMB 106 1.2D + 1.6W + 0.5L
156. 1 1.2 4 1.6 2 0.5
157. ********************************************************************************
158. * UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
159. LOAD COMB 201 D + F
160. 1 1.0 3 1.0
161. LOAD COMB 202 D + F + L
162. 1 1.0 2 1.0 3 1.0
163. LOAD COMB 203 D + F + W
164. 1 1.0 3 1.0 4 1.0
165. LOAD COMB 204 0.6D + W
166. 1 0.6 4 1.0
167. LOAD COMB 205 0.6D + 0.7E
168. 1 0.6 5 0.7
169. LOAD COMB 206 D + 0.75W + 0.75L
170. 1 1.0 4 0.75 2 0.75
171. ********************************************************************************
172. PERFORM ANALYSIS
```





**P R O B L E M  S T A T I S T I C S**

**NUMBER OF JOINTS 85 NUMBER OF MEMBERS 4**
**NUMBER OF PLATES 64 NUMBER OF SOLIDS 0**
**NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 81**

**SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER**

**ORIGINAL/FINAL BAND-WIDTH= 73/ 17/ 60 DOF**
**TOTAL PRIMARY LOAD CASES = 5, TOTAL DEGREES OF FREEDOM = 267**
**TOTAL LOAD COMBINATION CASES = 12 SO FAR.**
**SIZE OF STIFFNESS MATRIX = 17 DOUBLE KILO-WORDS**
**REQRD/AVAIL. DISK SPACE = 12.5/ 146183.9 MB**

```
173. LOAD LIST 101 TO 106
174. START CONCRETE DESIGN
```

```
175. CODE ACI
176. TRACK 2 ALL
177. FC 35000 MEMB 4 190 193 194
178. CLB 0.05 MEMB 4 190 193 194
179. CLS 0.05 ALL
180. CLT 0.05 MEMB 4 190 193 194
181. MAXMAIN 20 ALL
182. MINMAIN 20 ALL
183. MINSEC 10 ALL
184. DESIGN COLUMN 4 190 193 194
```





==========

**COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 20 MM | 1.005 | 101 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 8402.44 | 6721.95 | 3052.41 | 594.14 | 194.6 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 222.73 | -1039.71 | 21.54 | 0.00 | 0.00000 |

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 8402.44 | 6721.95 | 3052.41 | 594.14 | 194.6 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 222.73 | -1039.71 | 21.54 | 0.00 | 0.00000 |



**COLUMN NO. 190 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 20 MM | 1.005 | 101 | END | 0.650 |

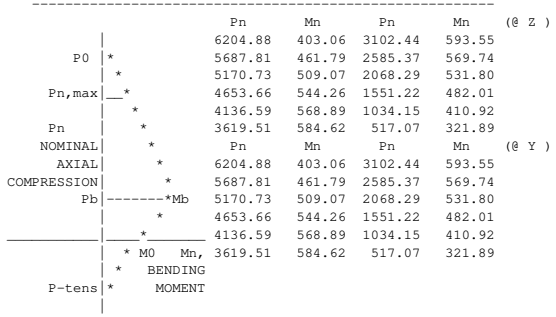
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 8402.44 | 6721.95 | 3052.41 | 594.14 | 194.6 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 222.73 | -1039.71 | 21.54 | 0.00 | 0.00000 |

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 8402.44 | 6721.95 | 3052.41 | 594.14 | 194.6 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 222.73 | -1039.71 | 21.54 | 0.00 | 0.00000 |








**COLUMN NO. 193 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 20 MM | 1.005 | 101 | END | 0.650 |

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 8402.44 | 6721.95 | 3052.41 | 594.14 | 194.6 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 222.73 | -1039.71 | 21.54 | 0.00 | 0.00000 |

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 8402.44 | 6721.95 | 3052.41 | 594.14 | 194.6 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 222.73 | -1039.71 | 21.54 | 0.00 | 0.00000 |

==========

**COLUMN NO. 194 DESIGN PER ACI 318-08 - AXIAL + BENDING**

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

| BAR CONFIGURATION | REINF PCT. | LOAD | LOCATION | PHI |
|---|---|---|---|---|
| 8 - 20 MM | 1.005 | 101 | END | 0.650 |

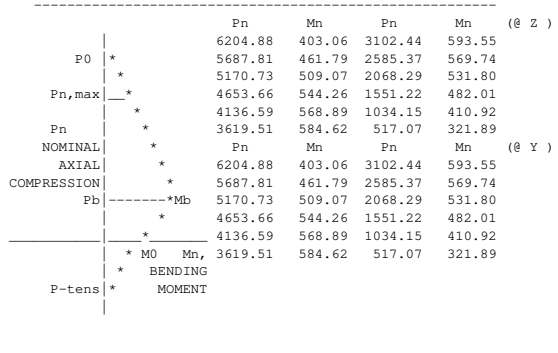
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 8402.44 | 6721.95 | 3052.41 | 594.14 | 194.6 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 222.73 | -1039.71 | 21.54 | 0.00 | 0.00000 |

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

| P0 | Pn max | P-bal. | M-bal. | e-bal. (MM) |
|---|---|---|---|---|
| 8402.44 | 6721.95 | 3052.41 | 594.14 | 194.6 |
| M0 | P-tens. | Des.Pn | Des.Mn | e/h |
| 222.73 | -1039.71 | 21.54 | 0.00 | 0.00000 |




**********END OF COLUMN DESIGN RESULTS**********





185. END CONCRETE DESIGN
186. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 26,2020 TIME= 16:49:55 ****

**********
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc. *
* http://www.bentley.com *
**********

```
            **************************************************
            *                                                *
            *          STAAD.Pro V8i SELECTseries5           *
            *          Version  20.07.10.64                  *
            *          Proprietary Program of                *
            *          Bentley Systems, Inc.                 *
            *          Date=    MAY 26, 2020                 *
            *          Time=    16:52: 3                     *
            *                                                *
            *     USER ID: LNT ECC DIV                       *
            **************************************************
```

```
     1. STAAD SPACE
INPUT FILE: 3650-COL-001 _002_STABILITY - NEW.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 18.05.2020
     4. JOB NAME MMGP/3650-COL-001 _002_STABILITY
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. CHECKER NAME SJM
    10. APPROVED NAME AG
    11. END JOB INFORMATION
    12. UNIT METER KN
    13. JOINT COORDINATES
    14. 7783 365.388 912.75 -383.293; 7787 365.388 911 -383.293
    15. 7968 366.675 911 -383.293; 7969 366.675 912.75 -383.293
    16. 7971 366.031 911 -382.65; 7972 366.031 912.75 -382.65
    17. 7973 366.031 911 -383.937; 7974 366.031 912.75 -383.937
    18. MEMBER INCIDENCES
    19. 4 7787 7783; 190 7968 7969; 193 7971 7972; 194 7973 7974
    20. DEFINE MATERIAL START
    21. ISOTROPIC CONCRETE
    22. E 2.17185E+007
    23. POISSON 0.17
    24. DENSITY 25
    25. ALPHA 1E-005
    26. DAMP 0.05
    27. TYPE CONCRETE
    28. STRENGTH FCU 27579
    29. END DEFINE MATERIAL
    30. MEMBER PROPERTY AMERICAN
    31. 4 190 193 194 PRIS YD 0.5 ZD 0.5
    32. CONSTANTS
    33. MATERIAL CONCRETE ALL
    34. SUPPORTS
    35. 7787 7968 7971 7973 FIXED
    36. ******************************************************************************
    37. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
```





```
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
           MASTER/SLAVE OR IF UNCONNECTED JOINTS.
    38. SELFWEIGHT Y -1
    39. JOINT LOAD
    40. 7783 7969 7972 7974 FY -10
    41. ******************************************************************************
    42. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
    43. JOINT LOAD
    44. 7783 7969 7972 7974 FY -50
    45. ******************************************************************************
    46. LOAD 3 LOADTYPE NONE  TITLE FLUID LOAD ( F )
    47. JOINT LOAD
    48. 7783 7969 7972 7974 FY -0.0001
    49. ******************************************************************************
    50. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD ( W )
    51. JOINT LOAD
    52. 7783 7969 7972 7974 FX 8 FY -100 FZ 8
    53. ******************************************************************************
    54. LOAD 5 LOADTYPE NONE  TITLE SEISMIC LOAD ( E )
    55. JOINT LOAD
    56. 7783 7969 7972 7974 FX 4 FY -57 FZ 4
    57. ******************************************************************************
    58. * UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE _STABILITY
    59. LOAD COMB 201 D + F
    60. 1 1.0 3 1.0
    61. LOAD COMB 202 D + F + L
    62. 1 1.0 2 1.0 3 1.0
    63. LOAD COMB 203 D + F + W
    64. 1 1.0 3 1.0 4 1.0
    65. LOAD COMB 204 0.6D + W
    66. 1 0.6 4 1.0
    67. LOAD COMB 205 0.6D + 0.7E
    68. 1 0.6 5 0.7
    69. LOAD COMB 206 D + 0.75W + 0.75L
    70. 1 1.0 4 0.75 2 0.75
    71. ******************************************************************************
    72. PERFORM ANALYSIS
```

```
        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS          8  NUMBER OF MEMBERS       4
  NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS      4


       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

   ORIGINAL/FINAL BAND-WIDTH=     1/     1/      6 DOF
 TOTAL      PRIMARY LOAD CASES =     5, TOTAL DEGREES OF FREEDOM =      24
 TOTAL LOAD COMBINATION  CASES =     6  SO FAR.
   SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
```

**REQRD/AVAIL. DISK SPACE = 12.0/ 146183.2 MB**

73. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 26,2020 TIME= 16:52: 7 ****

**********
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
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